

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 271 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

Long Short Neutral

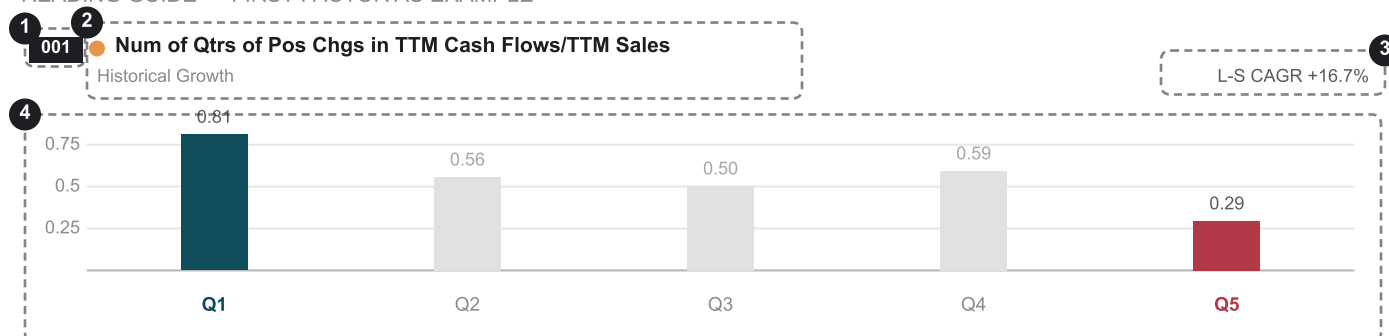
Long / short quintiles follow the live 48-month rules — spread-threshold based, not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

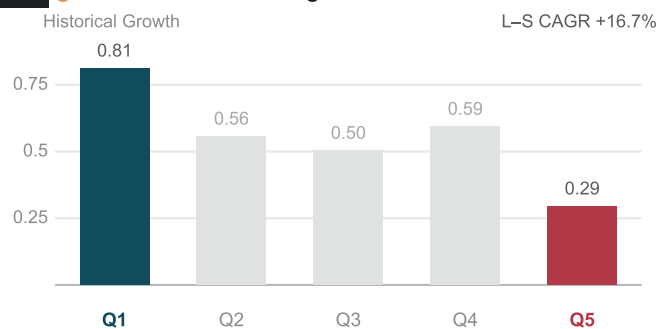
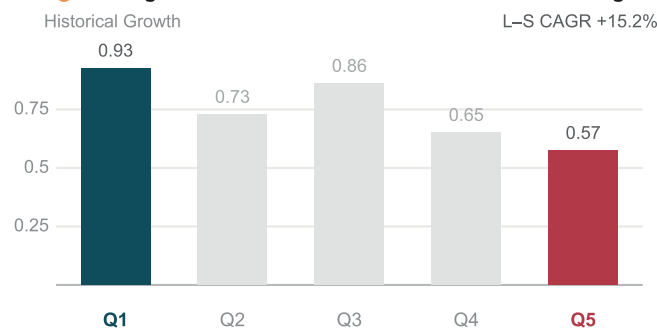
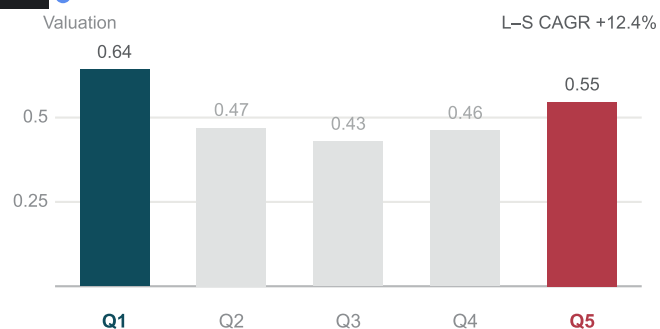
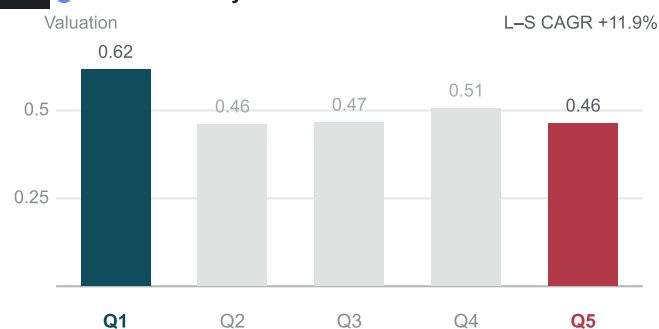
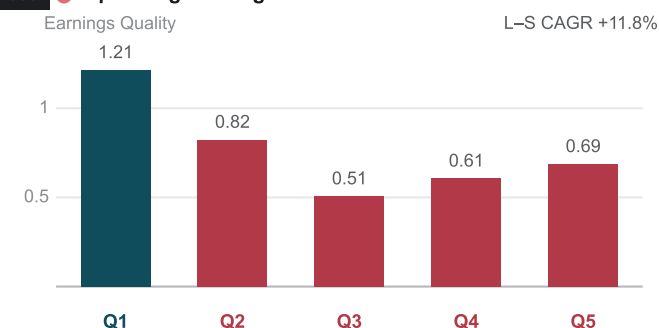
FACTOR STYLES

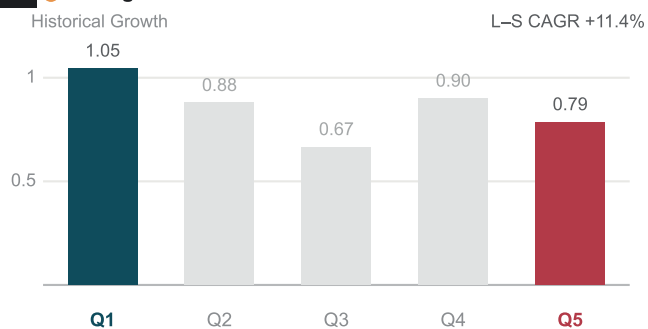
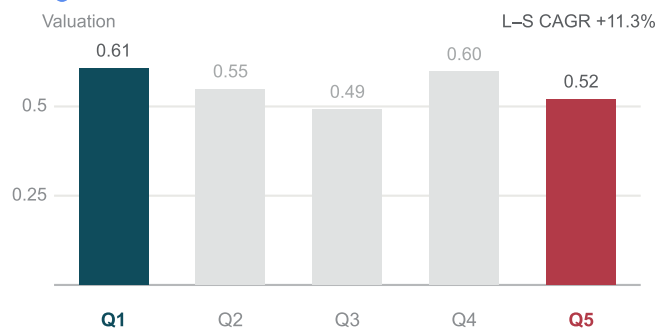
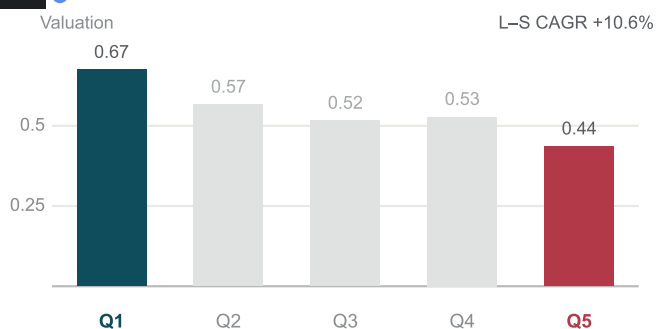
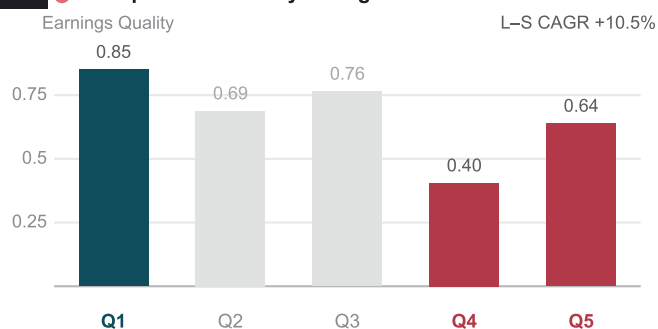
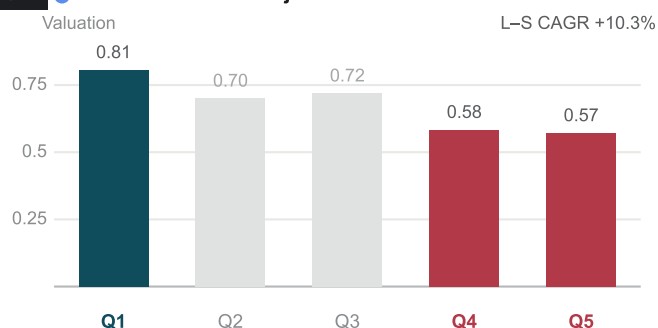
Valuation	41	Price Momentum	54
Earnings Quality	55	Size	3
Analyst Expectations	24	Historical Growth	56
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

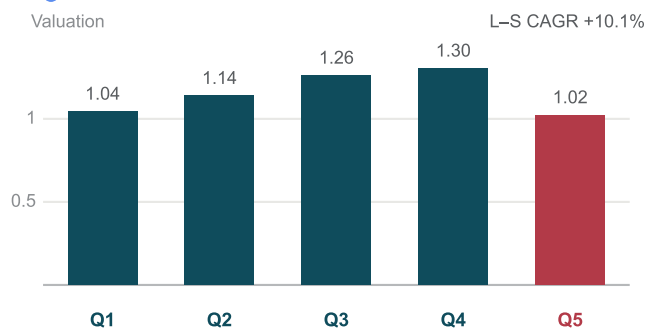
READING GUIDE — FIRST FACTOR AS EXAMPLE



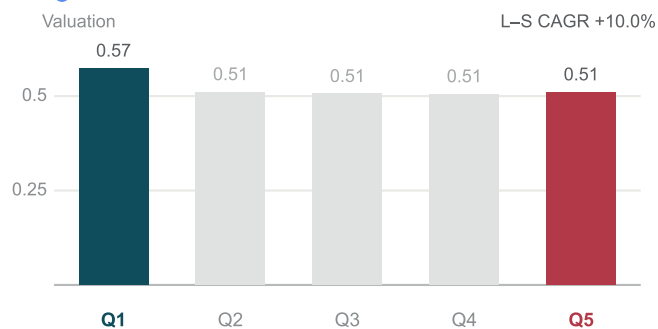
001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-M...****003 ● Pretax Income Per Share to Price****004 ● Current Liability to Price****005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flo****006 ● Operating Earnings to Assets Ratio**

007 ● Change of Asset Turnover**008 ● Assets to Price Ratio****009 ● Sales to Price Ratio****010 ● Unexpected Inventory Change****011 ● Log of Total Last Quarter Assets****012 ● Inverse P/E Ratio Adj for Growth and Yield**

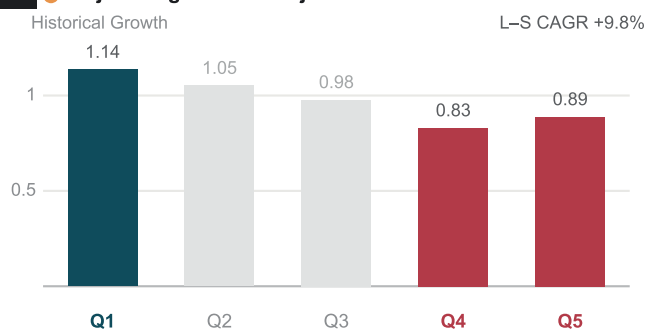
013 ● Indicated Annual Dividends to Price Ratio



014 ● Cash to Price



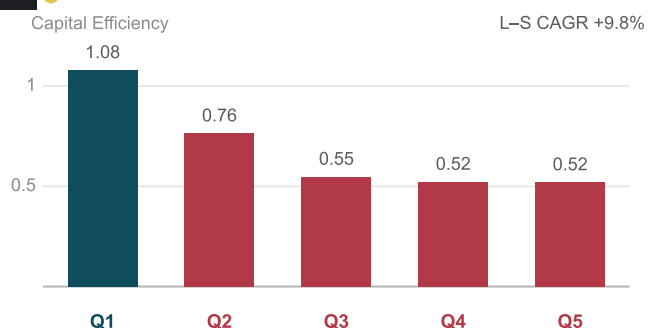
015 ● Adj 1Y Chg in Asset Adjusted FCF



016 ● Long Term Debt to Equity Ratio

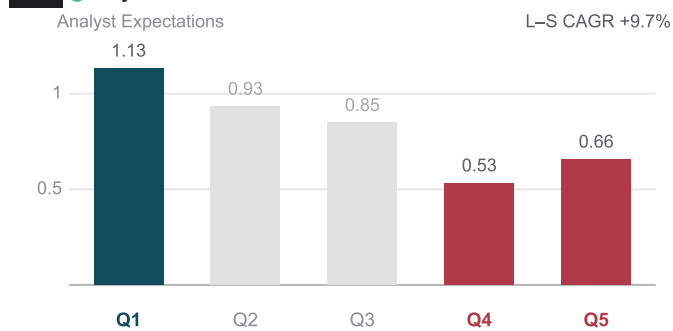
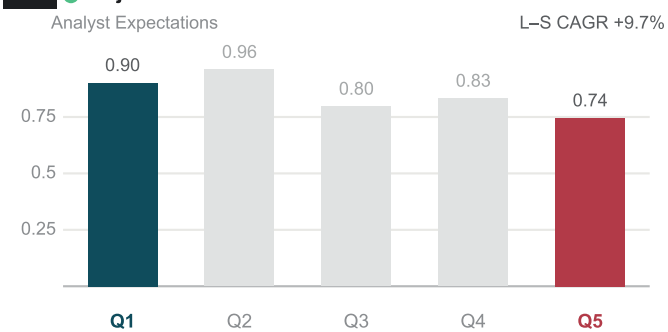
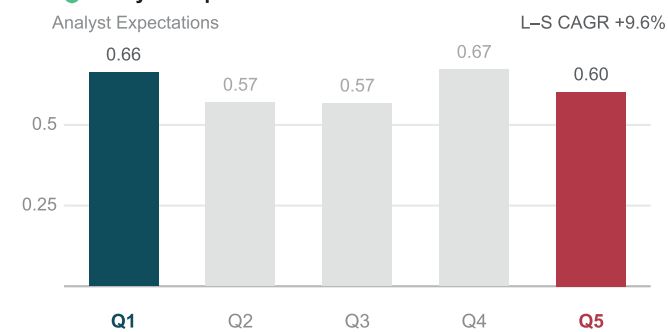
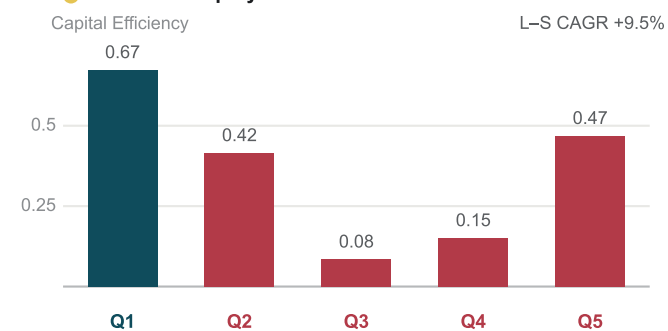


017 ● Return on Assets



018 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...

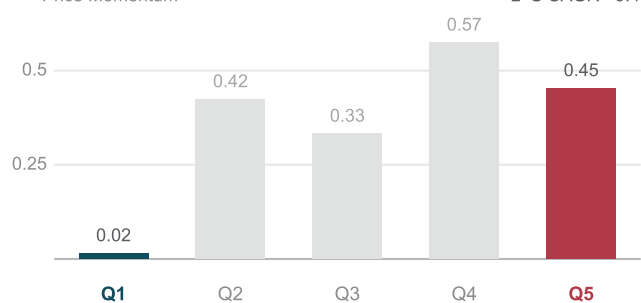


019 ● Gross Profit Margin**020 ● Adjusted 3M Revision in FY1 EPS Estimate****021 ● Adjusted Inverse PEG****022 ● Analyst Dispersion for FY2 EPS****023 ● Return on Equity****024 ● Forward Earnings to Price**

025 ● Sharpe Ratio

Price Momentum

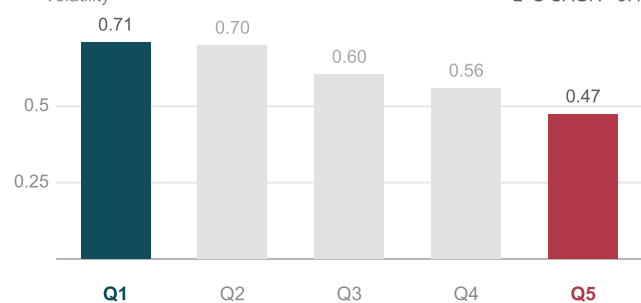
L-S CAGR +9.4%



026 ● 12M Realized Price Volatility

Volatility

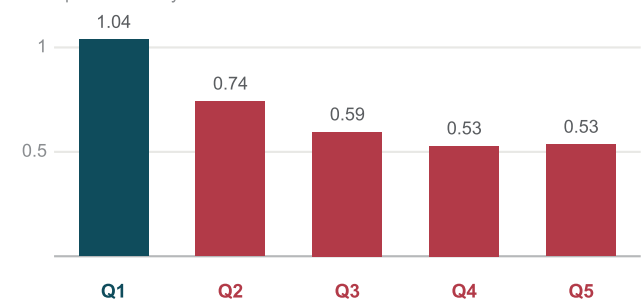
L-S CAGR +9.4%



027 ● Return on Invested Capital

Capital Efficiency

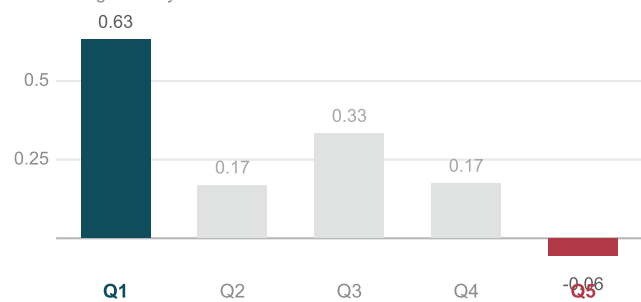
L-S CAGR +9.4%



028 ● Quick Ratio

Earnings Quality

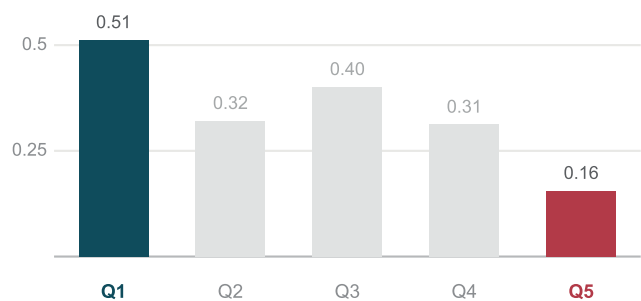
L-S CAGR +9.3%



029 ● 3Y Chg in Operating Cash Flow to Price

Historical Growth

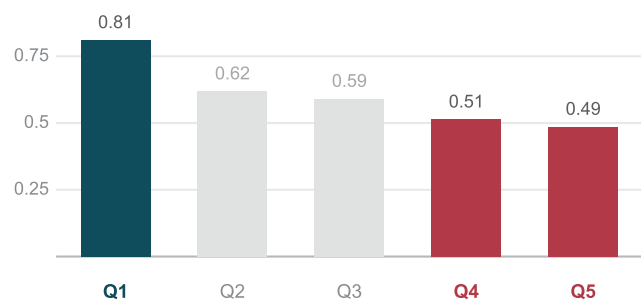
L-S CAGR +9.2%

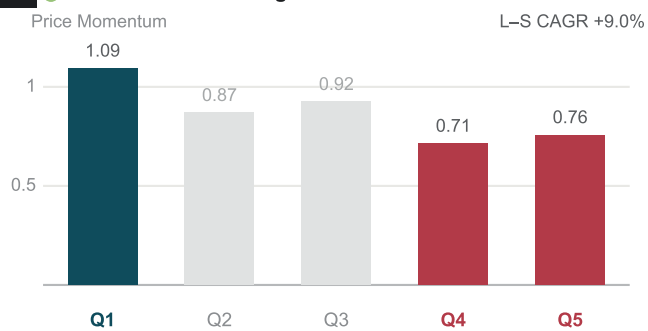
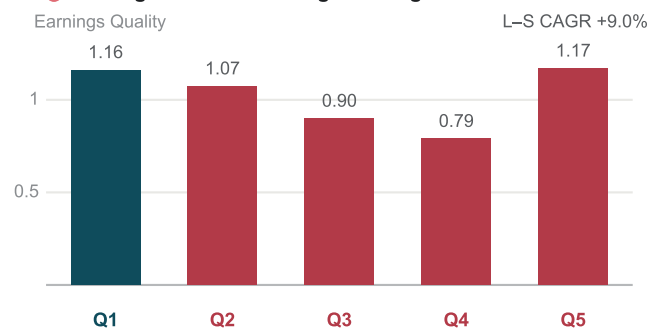
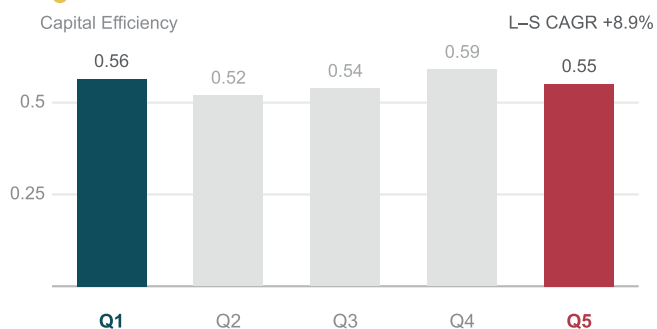
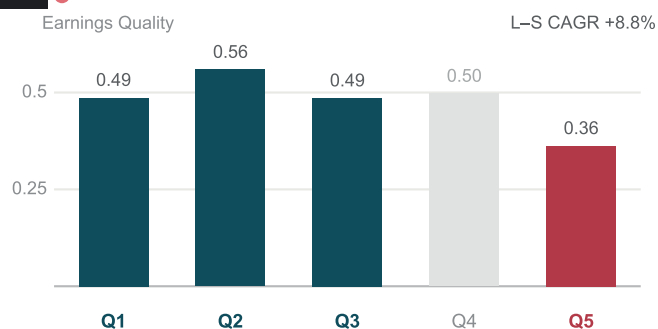
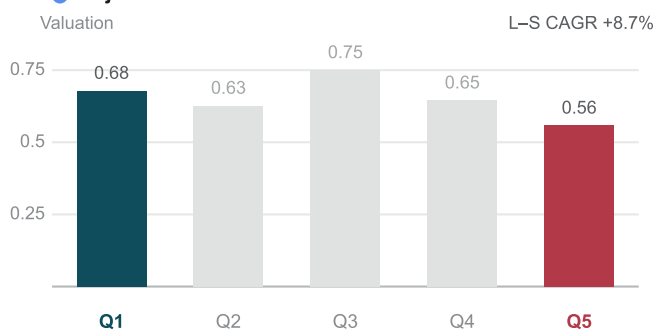


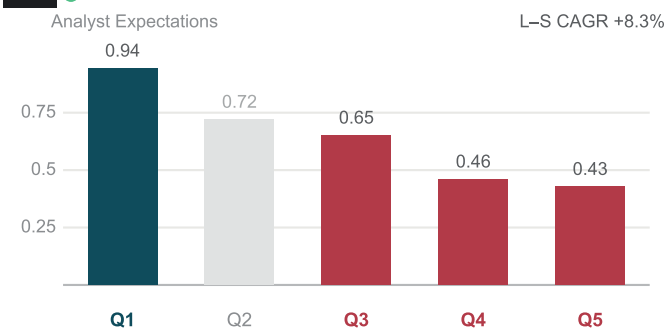
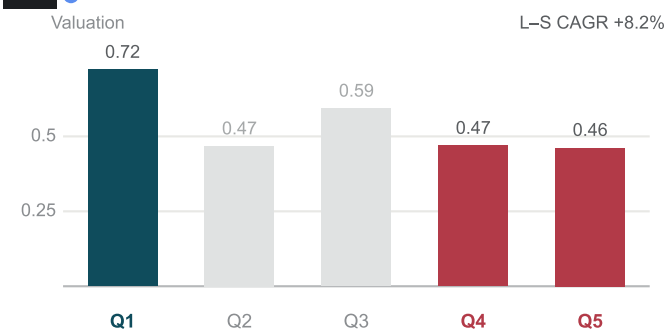
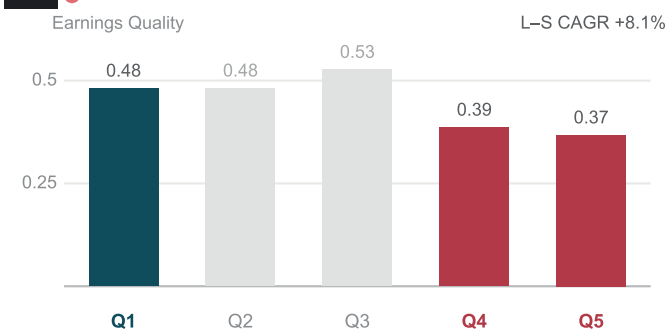
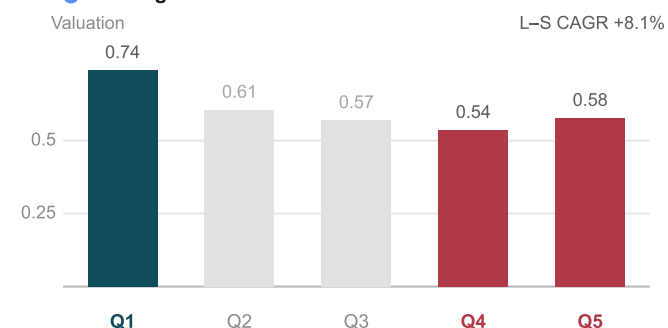
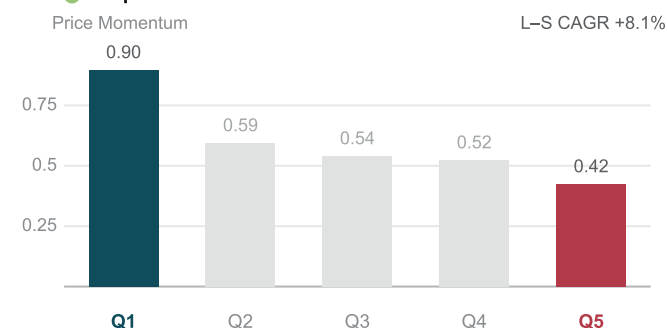
030 ● Operating Earnings to Price Ratio

Valuation

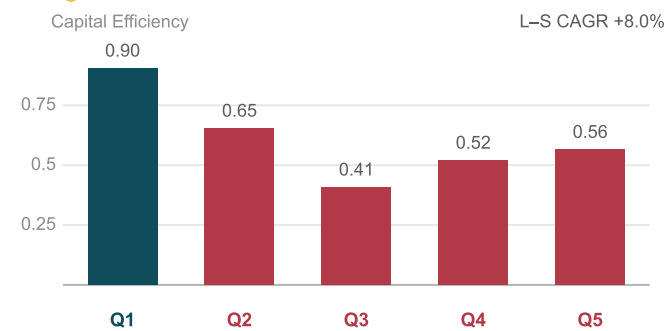
L-S CAGR +9.1%



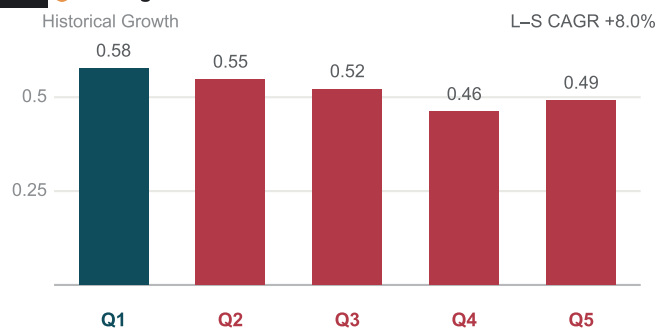
031 ● Short Interest Trading Volume**032 ● Trailing 12-Month Earnings/Trailing 12-Month Sales****033 ● Tobin's Q Ratio****034 ● ROA 20 Qtr Standard Deviation****035 ● Adj Growth Flow to Price****036 ● Working Capital to Sales**

037 ● 3M Revision in FY1 EPS Estimate**038 ● Adjusted Forward Earnings to Price****039 ● Normalized EP****040 ● Balance Sheet Accruals****041 ● Earnings to Price****042 ● Slope of 52W Price Trend**

043 ● EBITDA to Assets



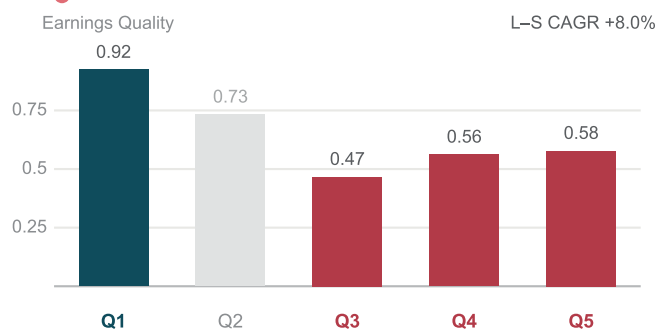
044 ● 3Y Chg in Cash Flow to Price



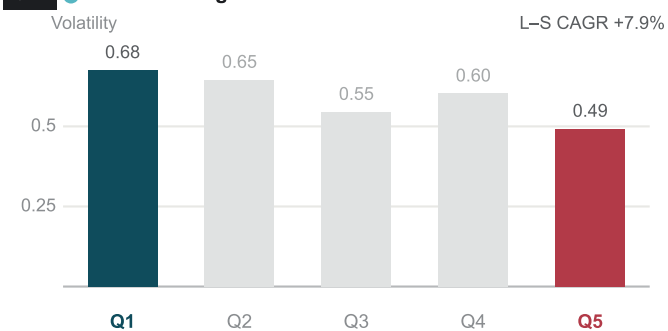
045 ● Consecutive Quarters of Declines in TTM Overhead/Sales



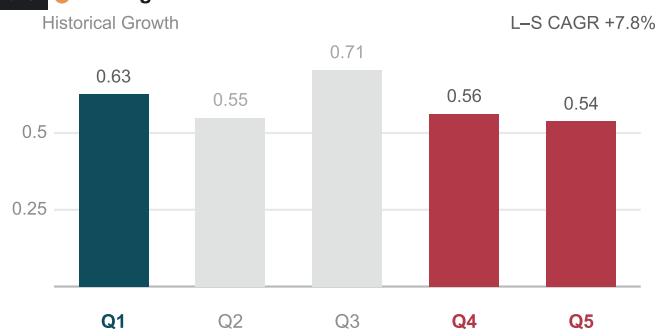
046 ● Cash Flow to Total Asset



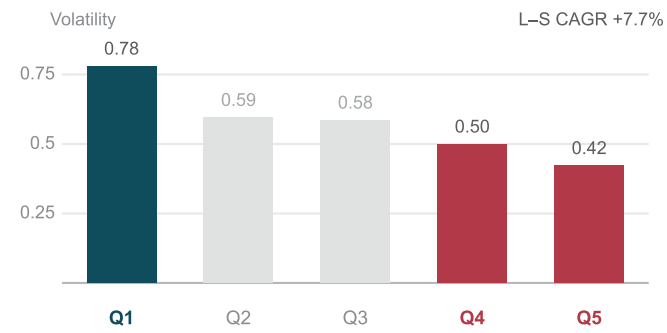
047 ● Book Leverage



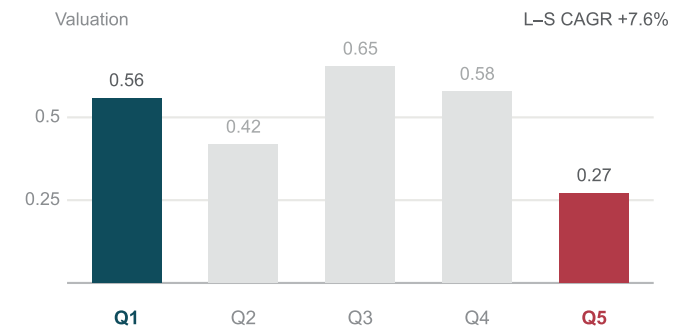
048 ● 1Y Chg in Sales to Price



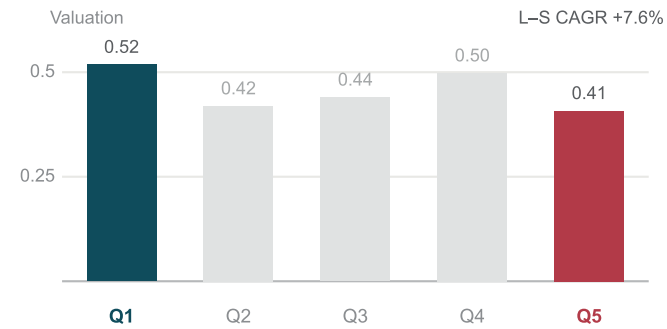
049 ● 60M CAPM Beta



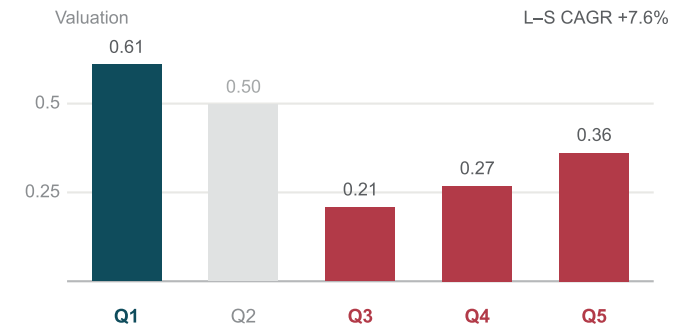
050 ● Adj Free Cash Flow to Price



051 ● Sales to EV Ratio



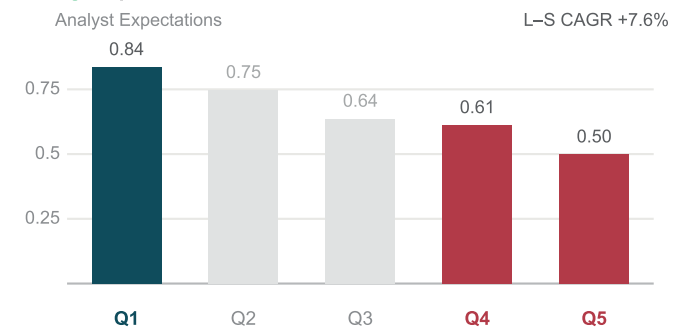
052 ● Cash Flow to Equity



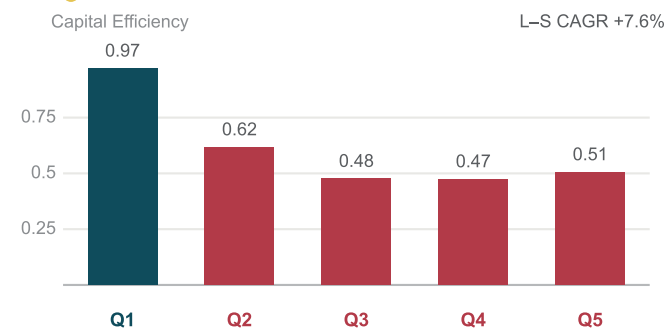
053 ● Cash to Adjusted Enterprise Value



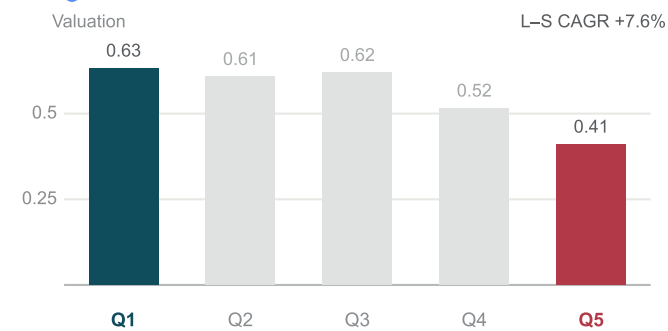
054 ● Expected LTG



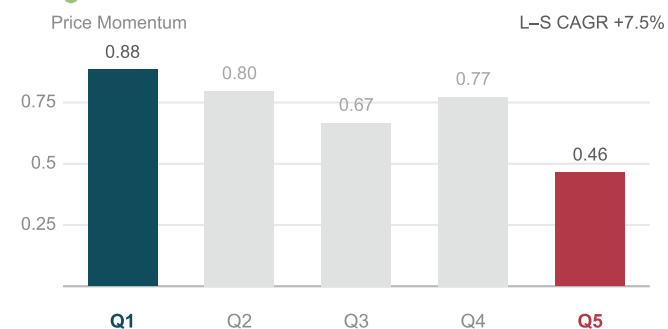
055 ● EBIT to Assets



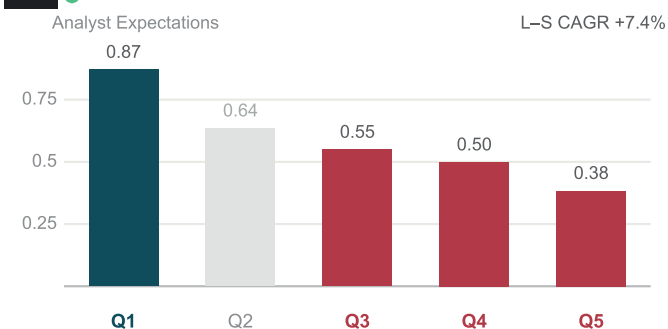
056 ● Dividends to Price Ratio



057 ● Short Interest Ratio



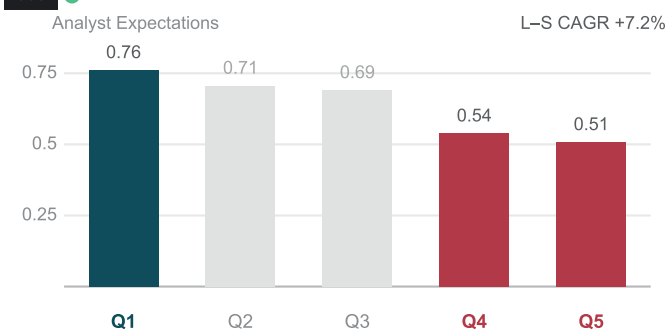
058 ● 3M Revision in FY2 EPS Estimate



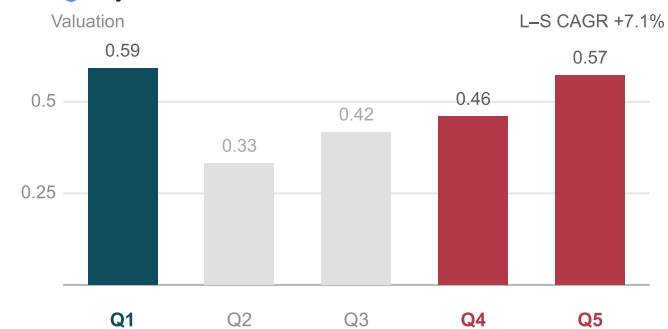
059 ● Amihud Illiquidity Measure



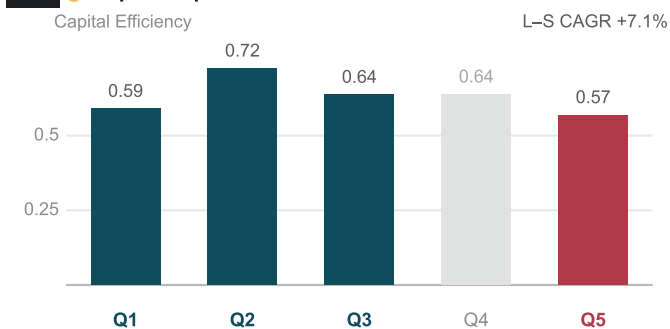
060 ● Inverse PEG



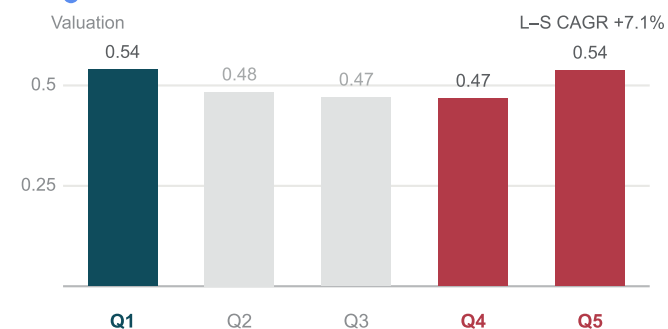
061 ● Adjusted EBIT to Price



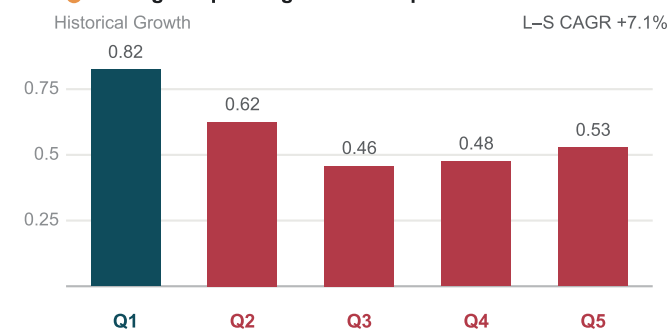
062 ● Capital Expenditure to Total Assets



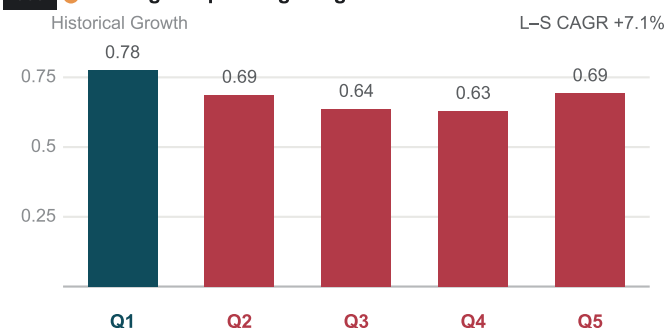
063 ● Net Cash Flow to Price



064 ● 3Y Chg in Operating Cash Flow per Share



065 ● 1Y Chg in Operating Margin



066 ● 1Y Chg in Cash Flow to Price



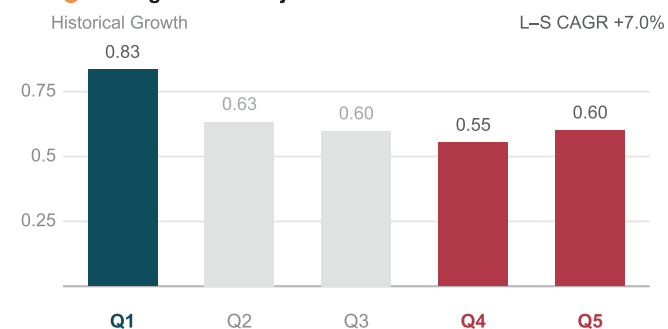
067 ● 50 Day to 200 Day Stock Price



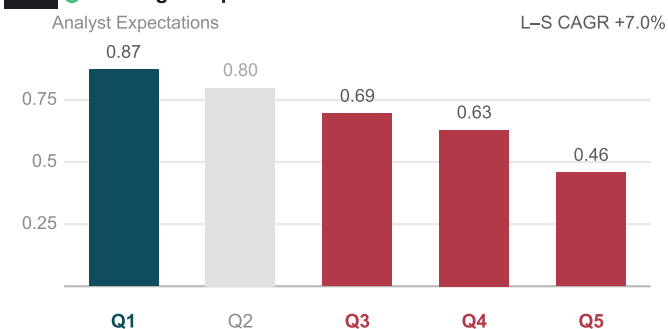
068 ● Adjusted Book to Enterprise Value



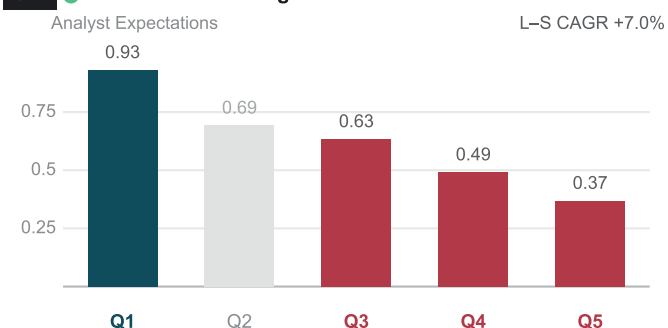
069 ● 1Y Chg in Asset Adjusted EPS



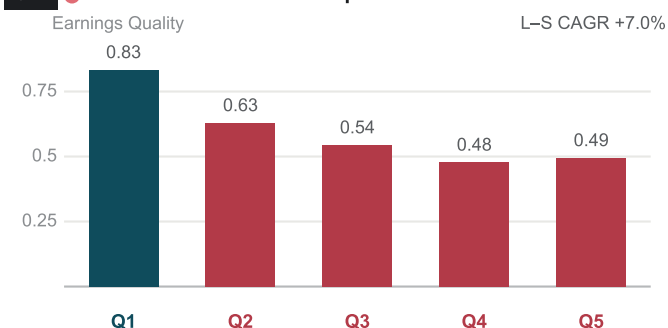
070 ● Earnings Surprise

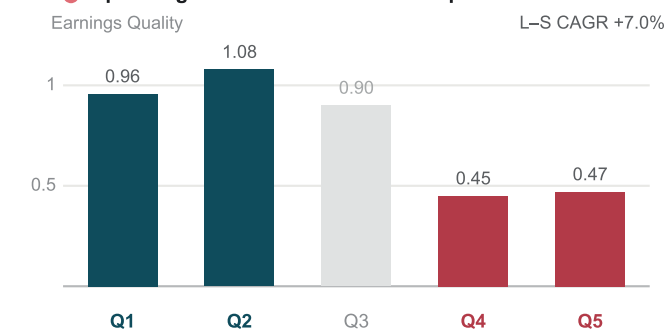
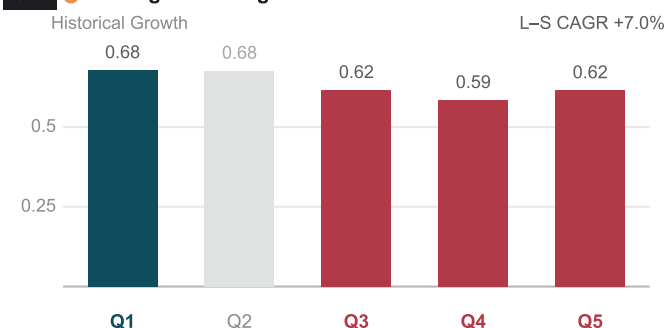
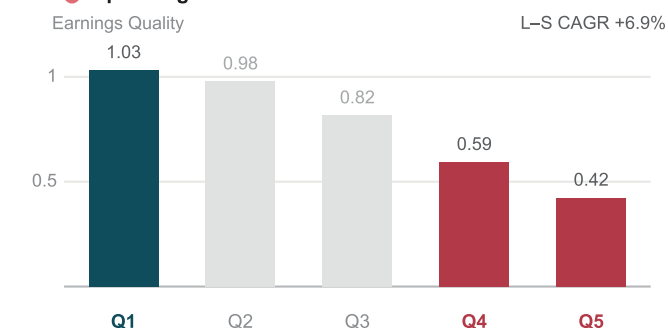


071 ● Street Revision Magnitude



072 ● Cash Flow to Invested Capital

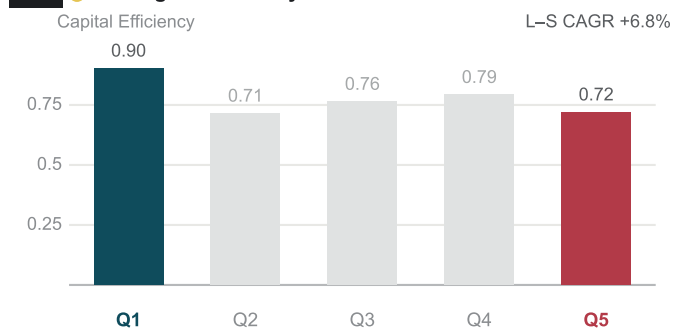


073 ● Sales to Invested Capital**074 ● 39W Lagged Return****075 ● Operating Cash Flow to Invested Capital****076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow****077 ● 1Y Chg in Earnings to Price****078 ● Operating Cash Flow to Asset**

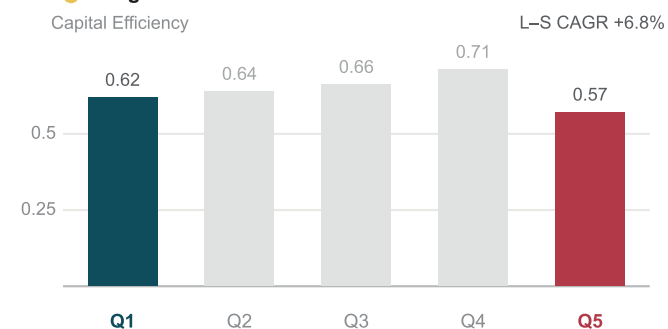
079 ● EBITDA to Enterprise Value



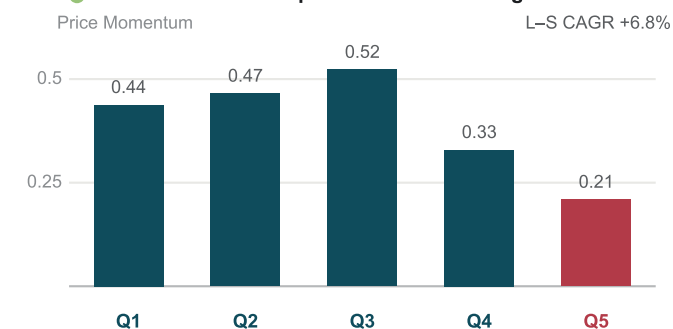
080 ● 1Y Chg in Stock Buybacks



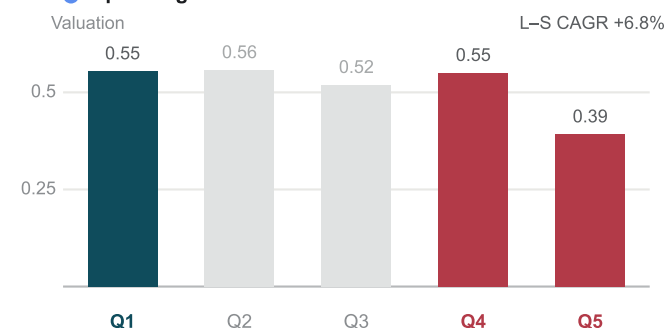
081 ● Long Term Debt to Assets Ratio



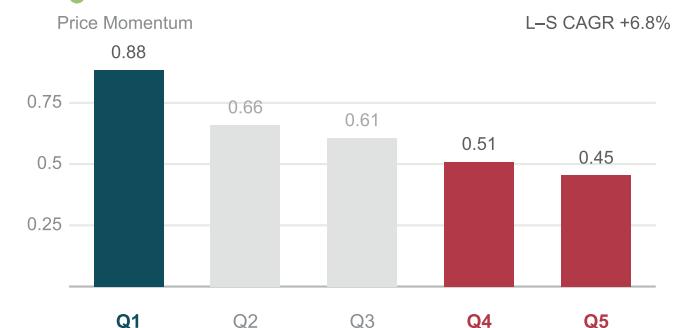
082 ● Institution Ownership Level - Active Manager

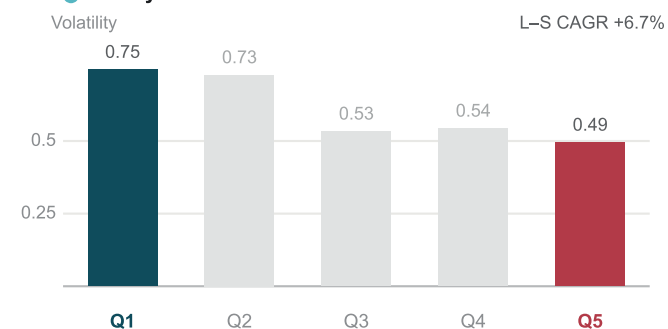
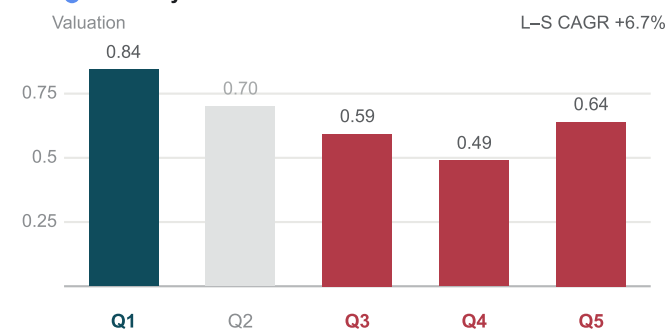
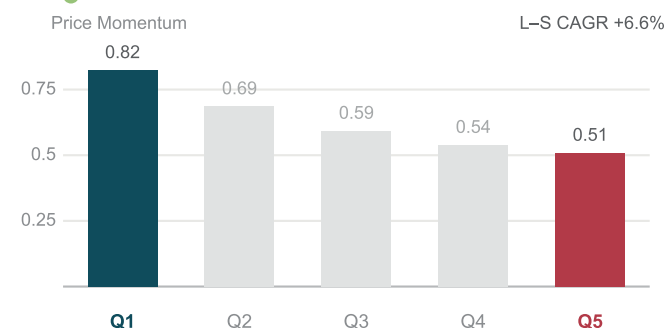
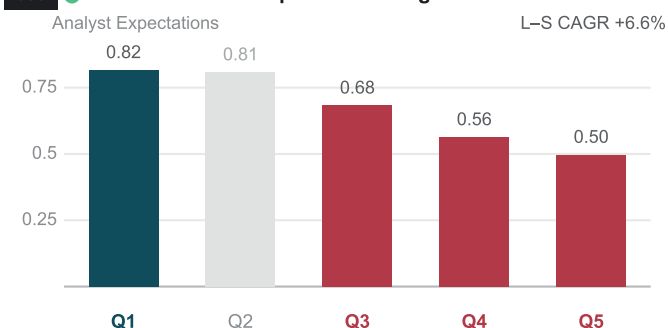


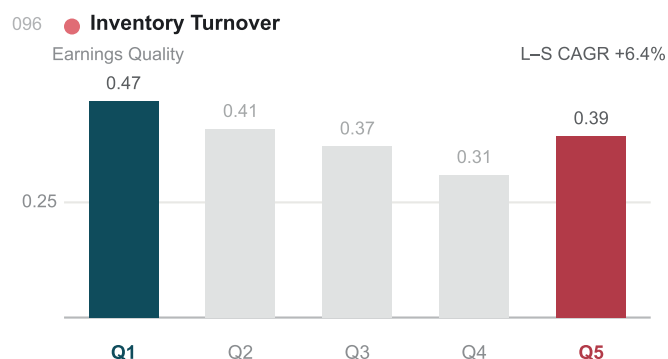
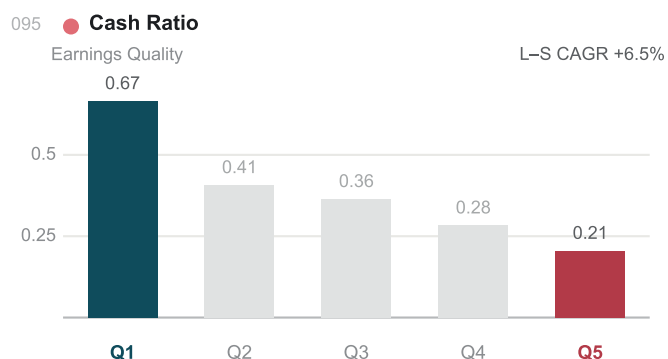
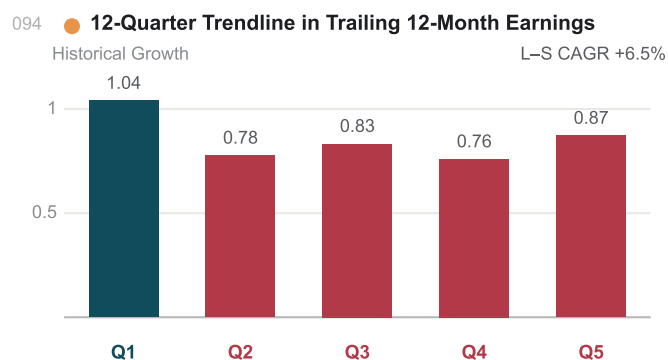
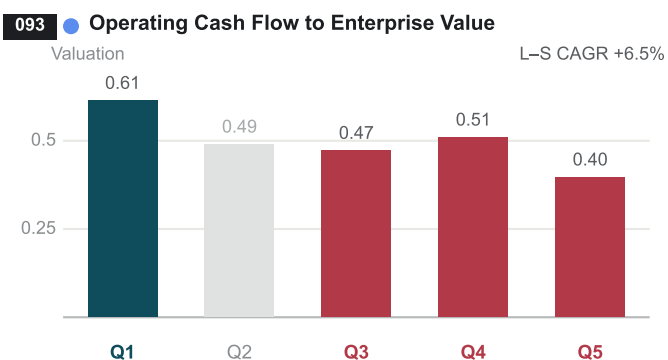
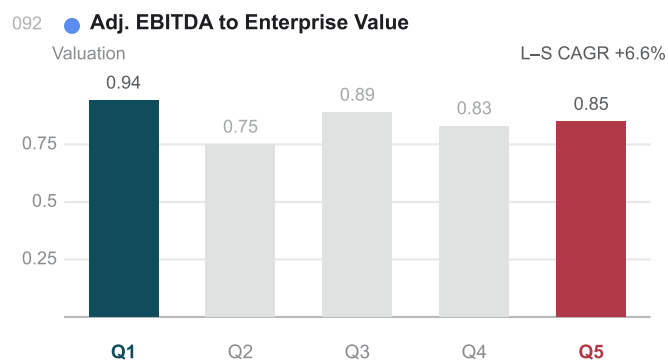
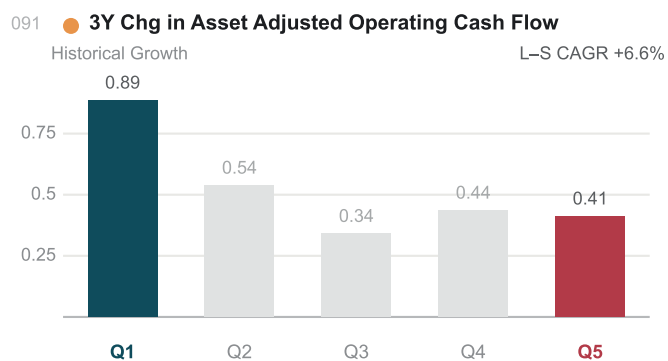
083 ● Operating Cash Flow to Price

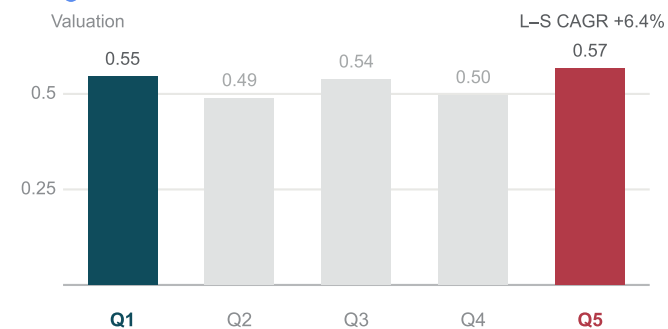
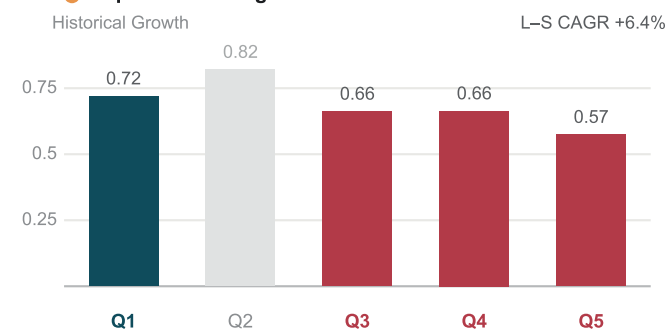
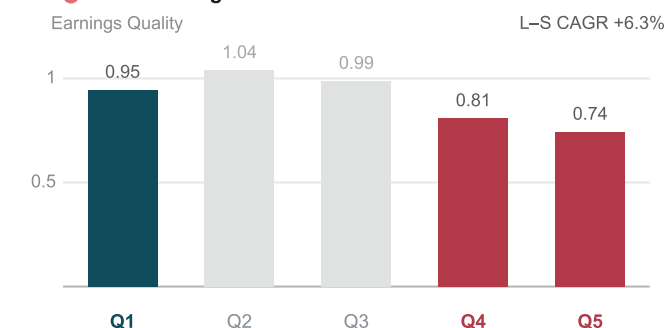
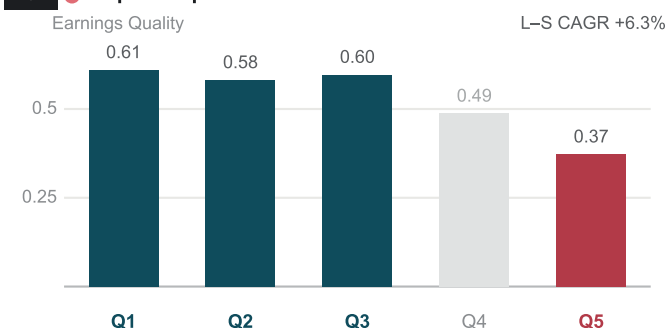


084 ● 15/36 Week Price Ratio

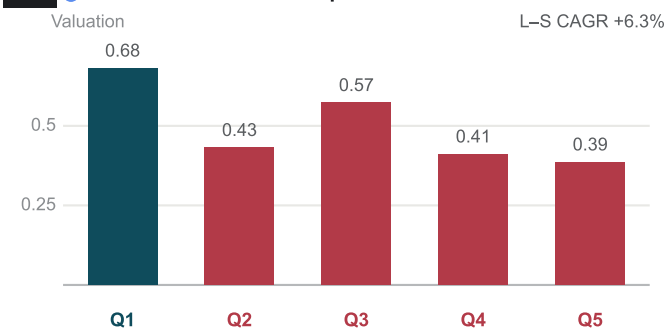


085 ● **26W Relative Price Strength**086 ● **Adj 1Y Chg in Earnings per Share**087 ● **90 Day Coefficient of Variation**088 ● **Solvency Ratio**089 ● **9 Month Price Momentum**090 ● **Standardized Unexpected Earnings**

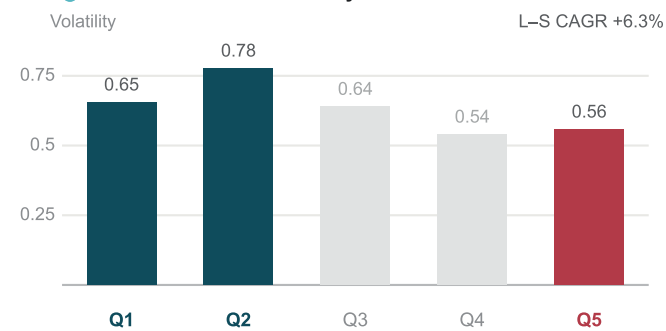


097 ● **Book to Price**098 ● **6M Chg in Target Price**099 ● **Adjusted Revision Magnitude**100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**101 ● **EBITDA Margin**102 ● **Capital Expenditure to Sales**

103 ● Free Cash Flow to Enterprise Value



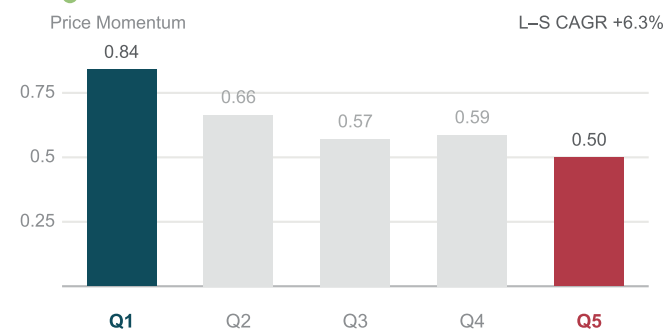
104 ● 1M Realized Price Volatility



105 ● EBITDA to Price



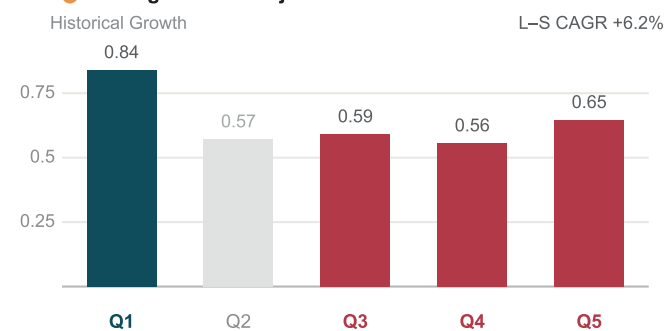
106 ● 4W to 52W Price Oscillator



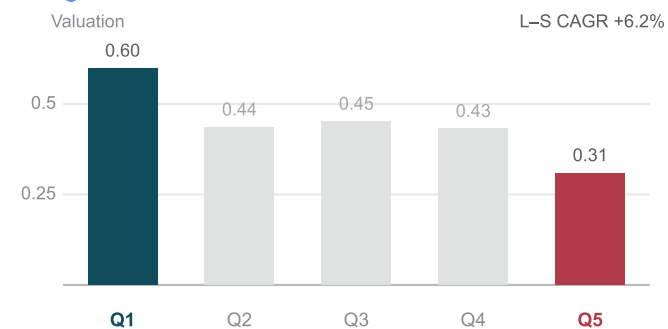
107 ● Percent Above 260-Day Low



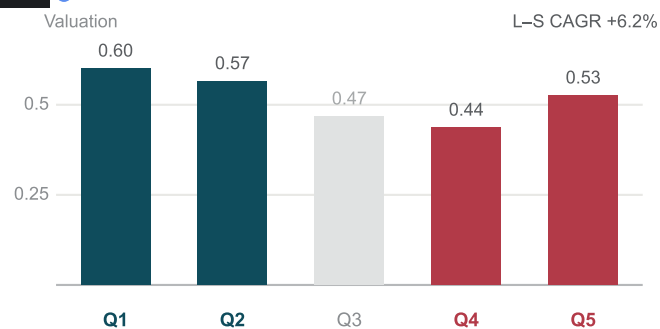
108 ● 1Y Chg in Asset Adjusted Cash Flow



109 ● Growth Flow to Price



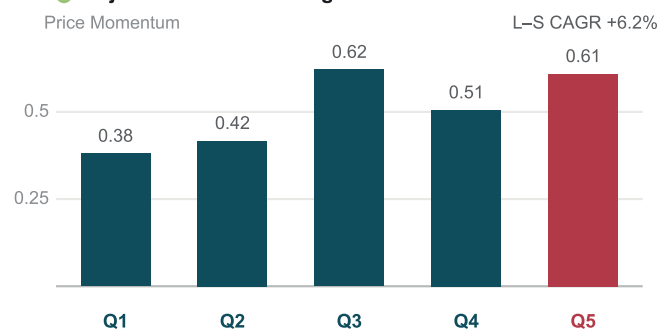
110 ● Cash to EV Ratio



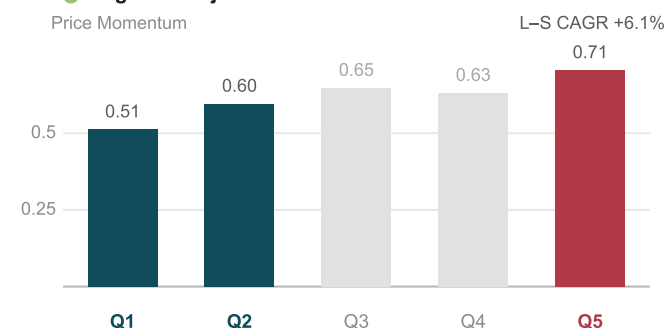
111 ● Year over Year Change in SGA to Sales



112 ● Adjusted 6 Month Average Share Turn Over



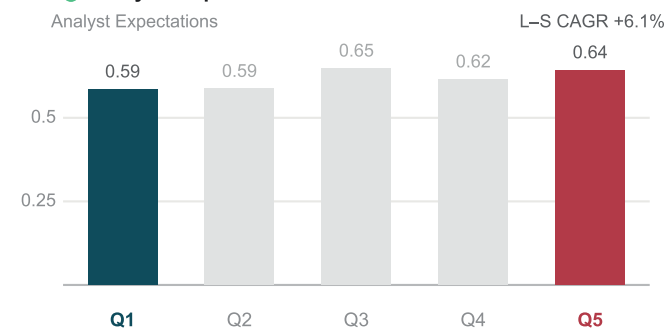
113 ● Log of Unadjusted Stock Price



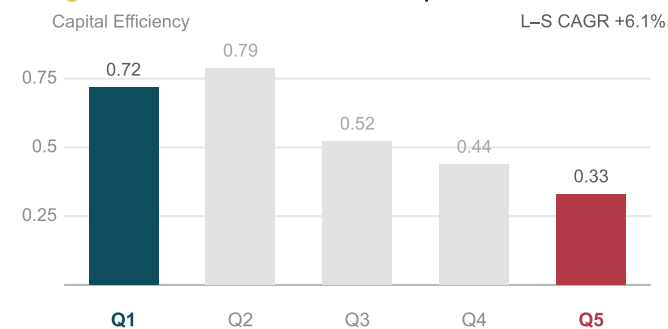
114 ● 1Y Chg in Operating Cash Flow per Share



115 ● Analyst Dispersion for FY1 EPS



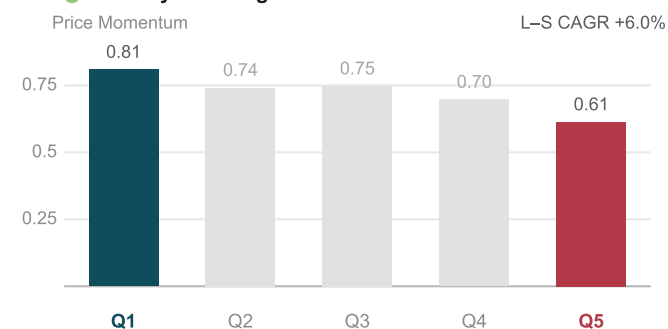
116 ● Cash Flow Return on Invested Capital



117 ● 1Y Chg in Asset Adjusted Free Cash Flow



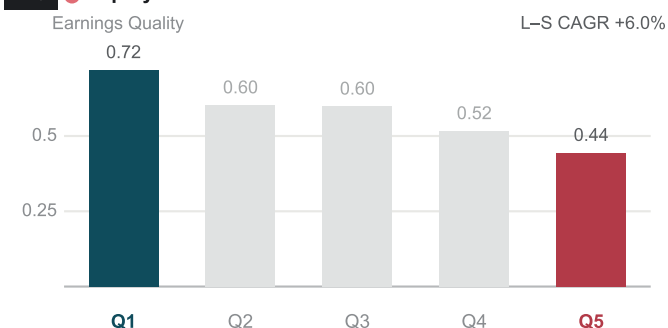
118 ● Stability of Change in Number of Holders

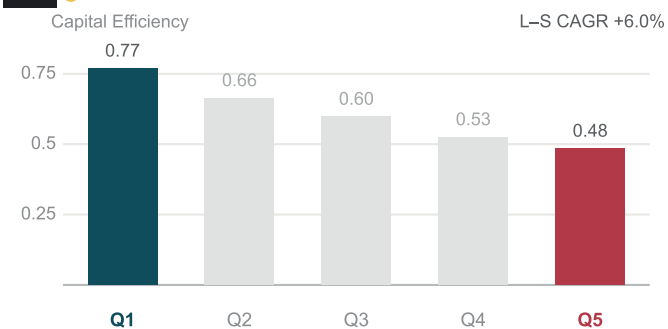
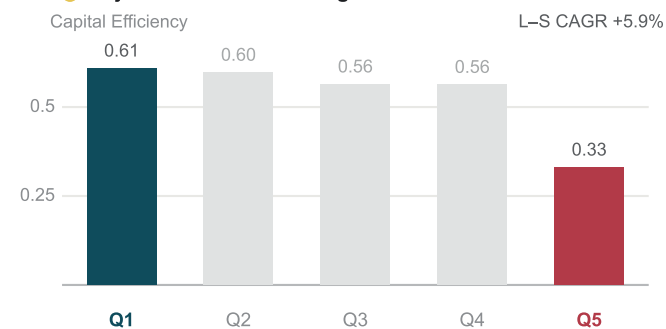
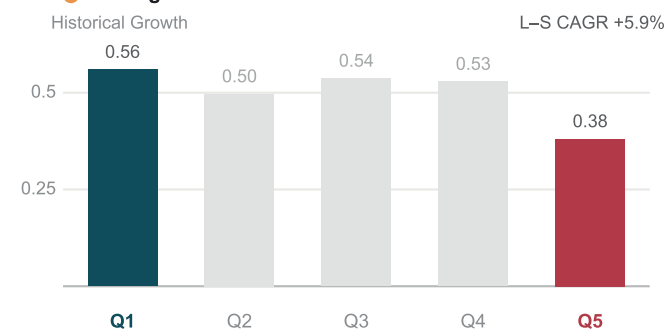
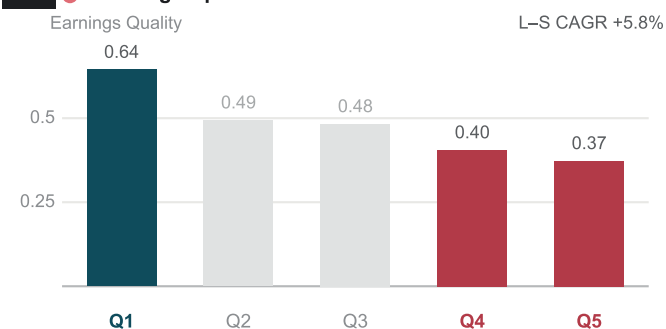
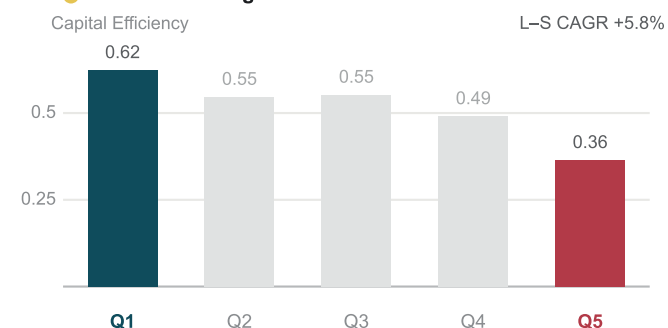
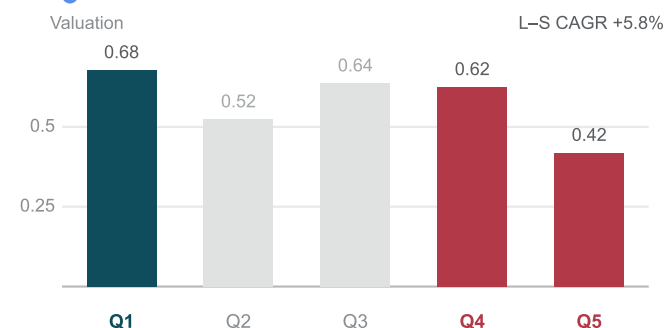


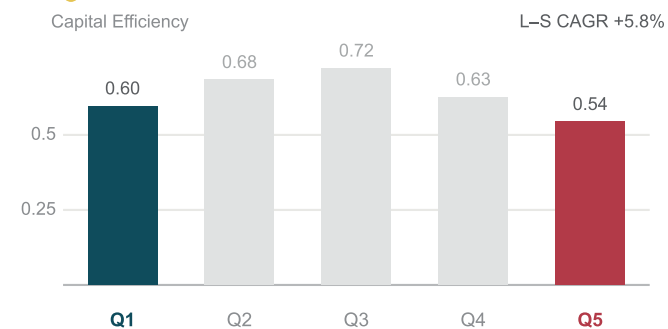
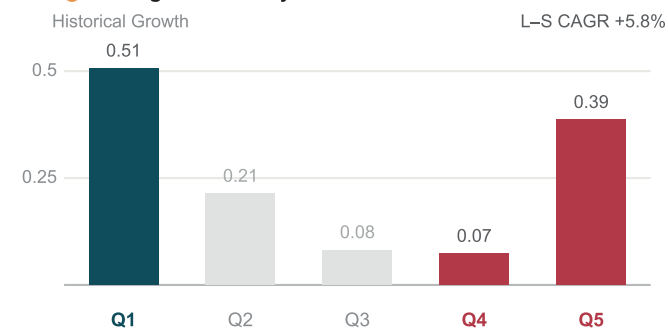
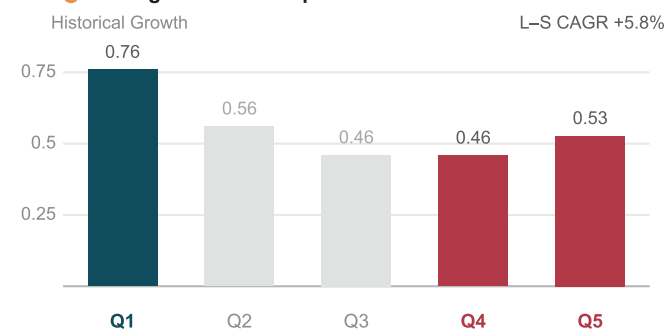
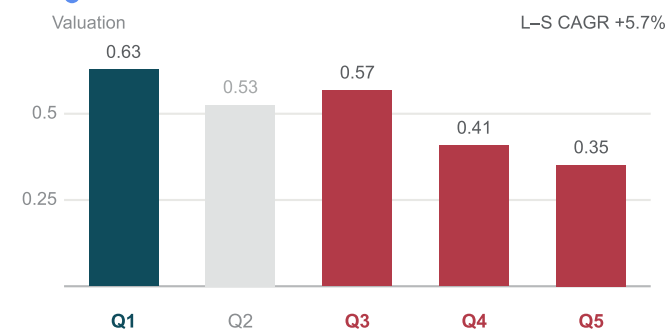
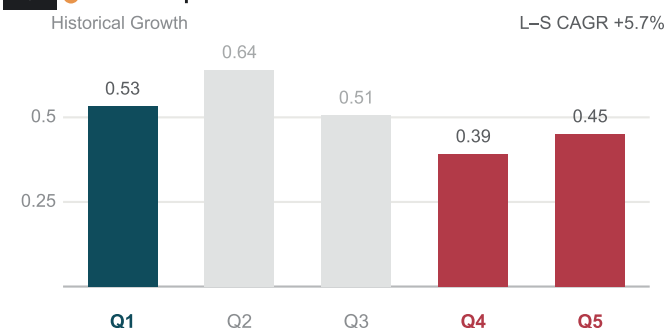
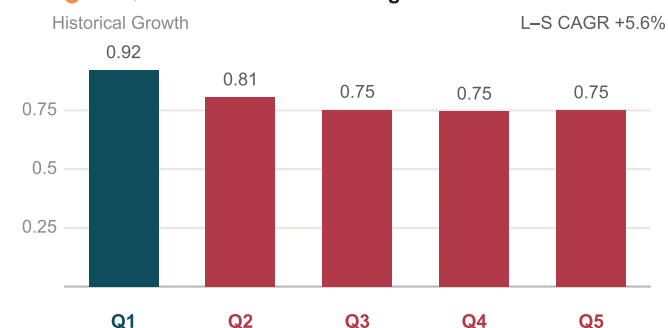
119 ● Operating Cash Flow Ratio



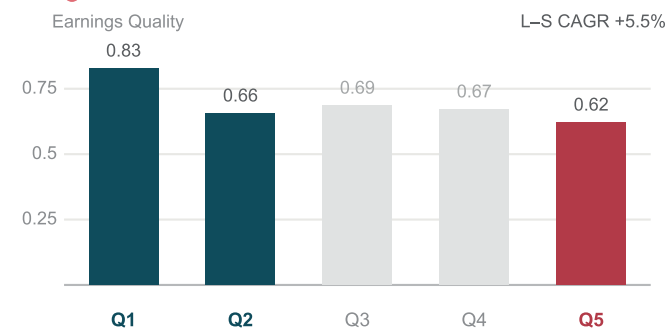
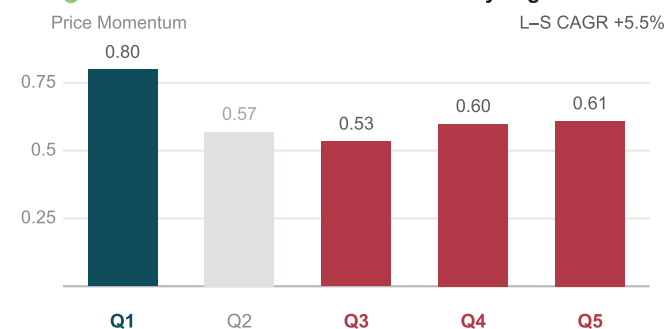
120 ● Equity Turnover

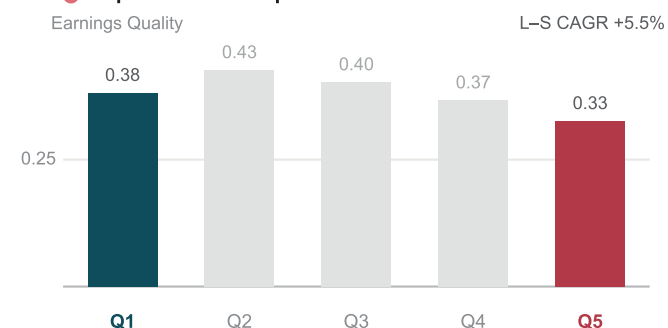
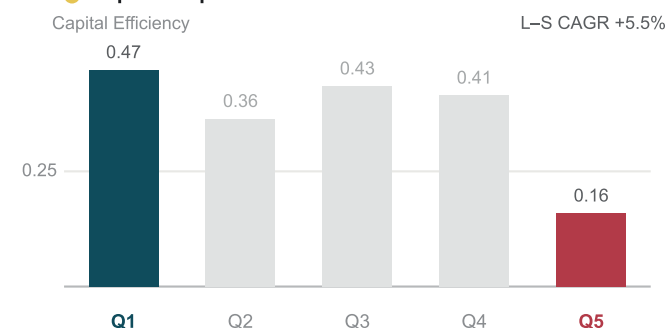


121 ● Receivables Turnover Ratio**122 ● Adjusted Interest Coverage Ratio****123 ● 3Y Chg in Free Cash Flow to Price****124 ● Working Capital Accruals****125 ● Interest Coverage Ratio****126 ● Forward Free Cash Flow to Price Ratio**

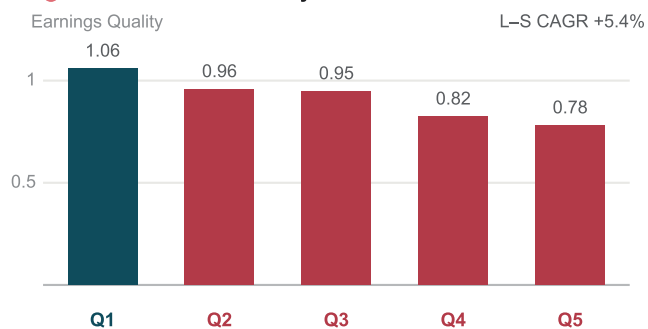
127 ● **Debt to Assets Ratio**128 ● **3Y Chg in Asset Adjusted Cash Flow**129 ● **3Y Chg in Cash Flow per Share**130 ● **Free Cash Flow to Price**131 ● **Dividend per Share Growth**132 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

133 ● 1Y Chg in Operating Cash Flow to Price

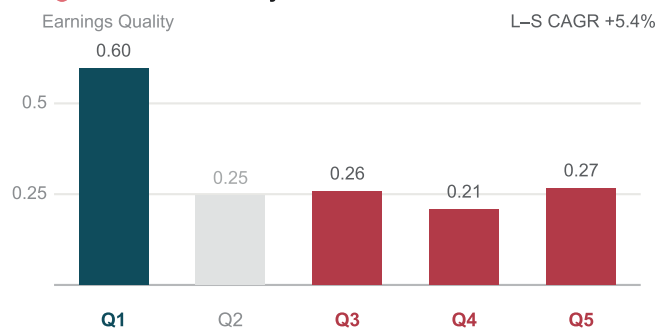
134 ● Cash Burn Rate

135 ● 52-Week Volume Price Trend with 20-Day Lag

136 ● 3Y Average Annual Earnings Growth

137 ● Depreciation to Capex Ratio

138 ● Capital Acquisition Ratio


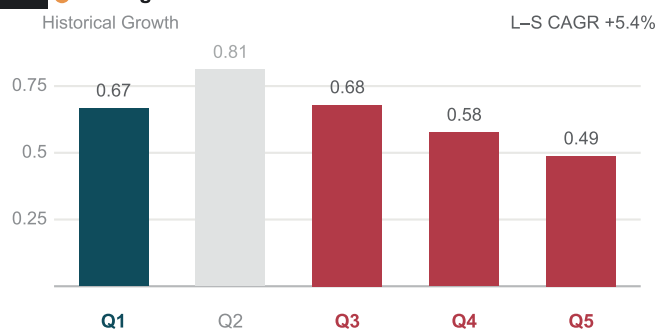
139 ● Free Cash Flow Stability



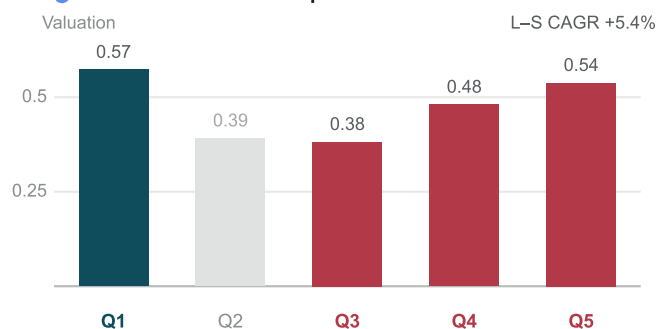
140 ● Net Income Stability



141 ● 1Y Chg in Free Cash Flow to Price



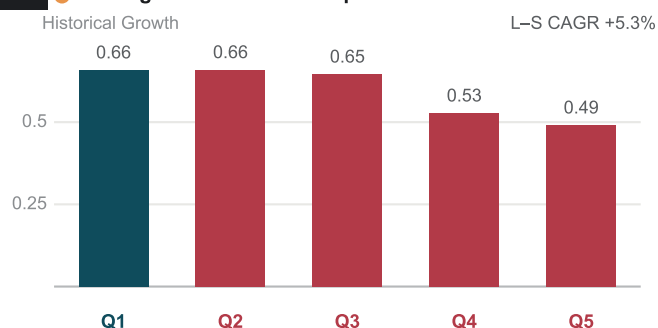
142 ● Net Cash Flow to Enterprise Value

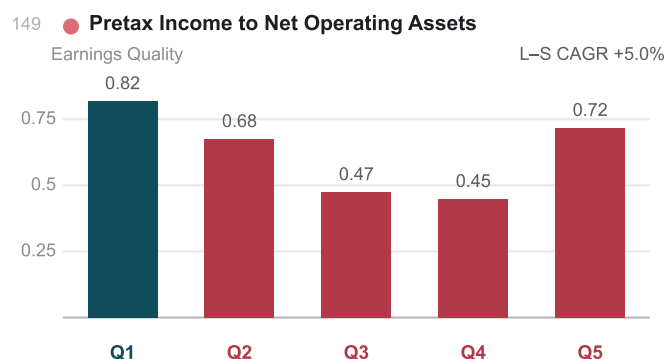
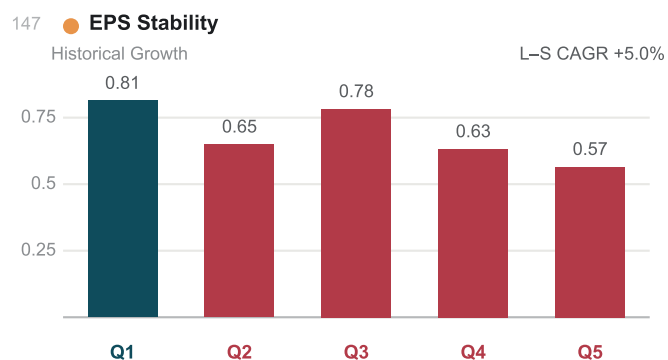
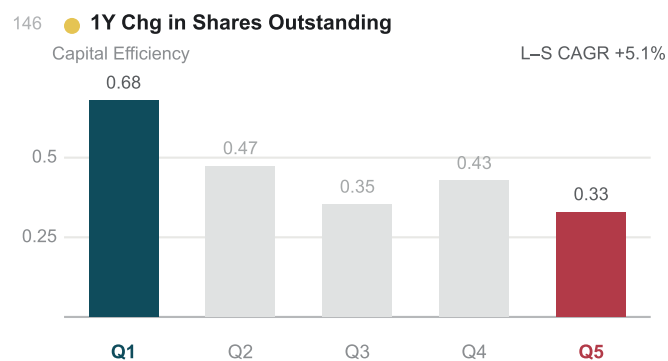
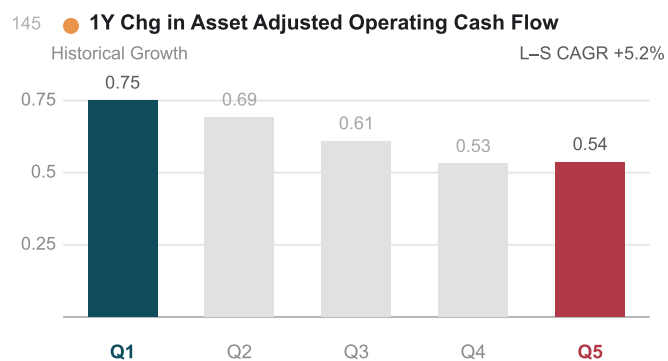


143 ● 12M - 1M Price Momentum

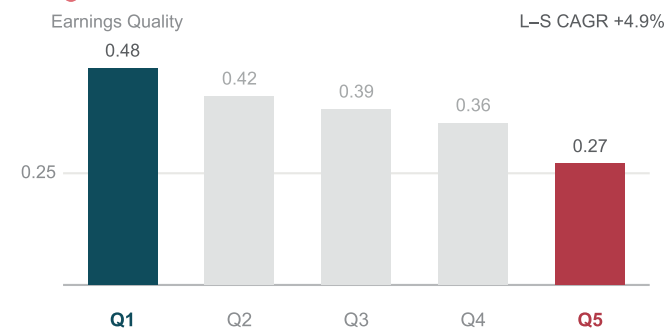


144 ● 1Y Chg in Free Cash Flow per Share

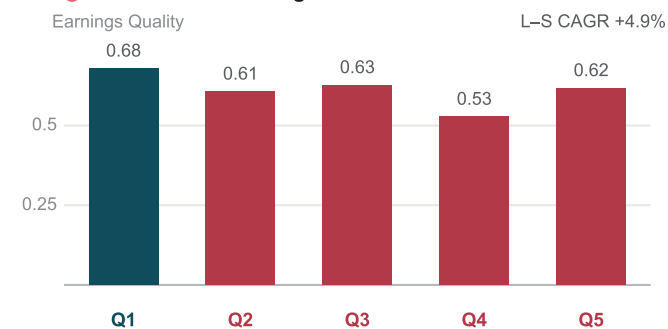




151 ● Free Cash Flow to Sales



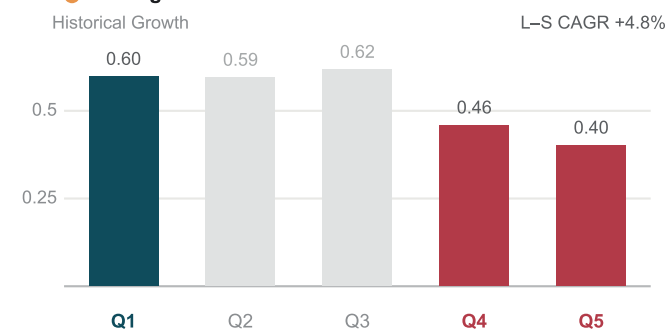
152 ● Year over Year Change of EPS to Sales



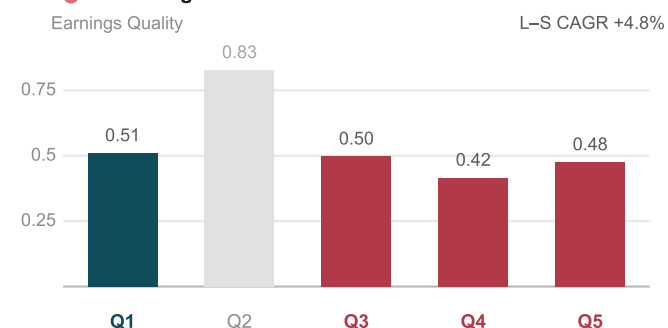
153 ● Operating Margin



154 ● 3Y Chg in Sales to Price



155 ● EBIT Margin



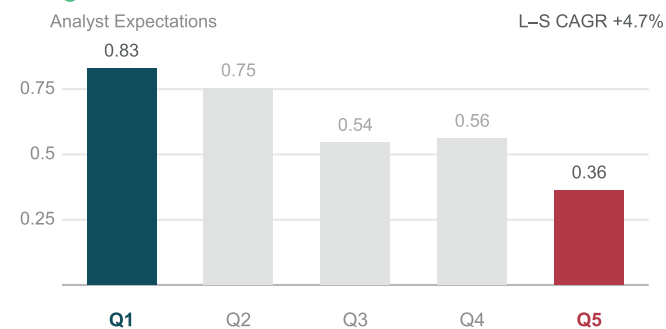
156 ● Adj Forward Free Cash Flow to Price Ratio



157 ● 3Y Chg in Free Cash Flow per Share



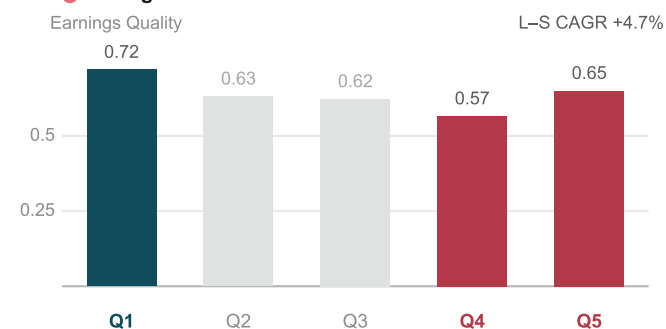
158 ● Number of EPS FY1 Revisions



159 ● Working Capital to Assets Ratio



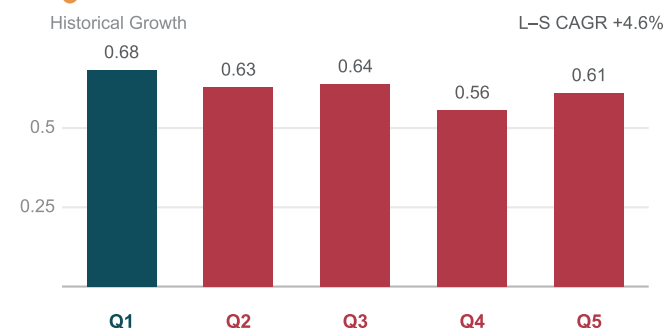
160 ● Change EPS to Sales

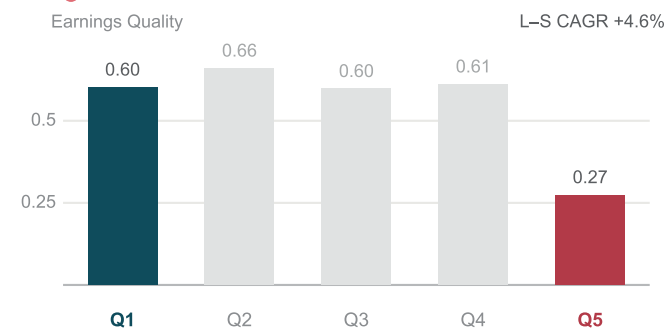
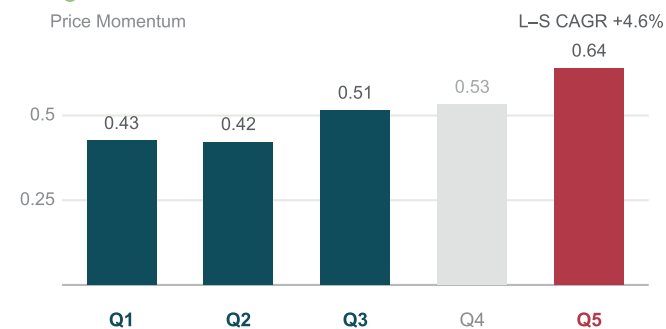
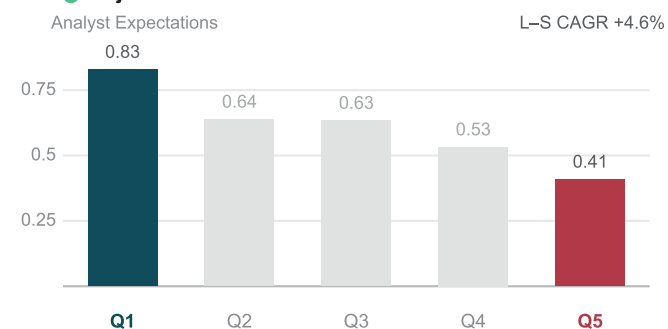
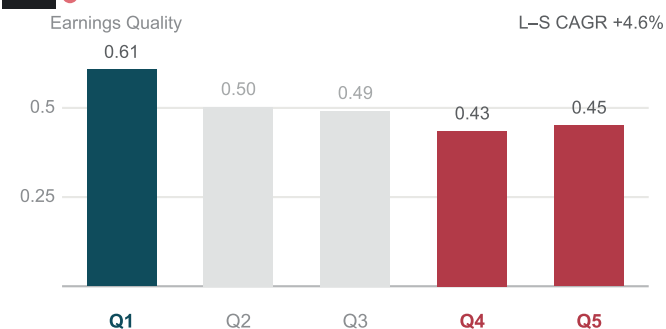
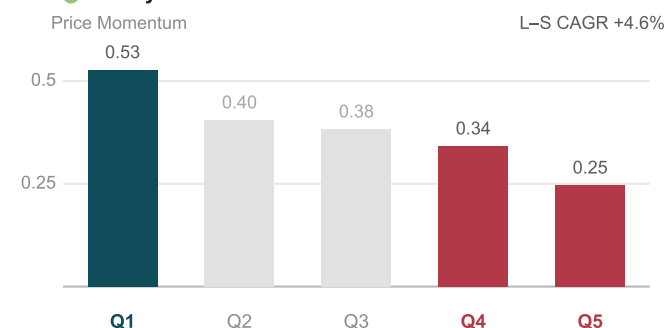
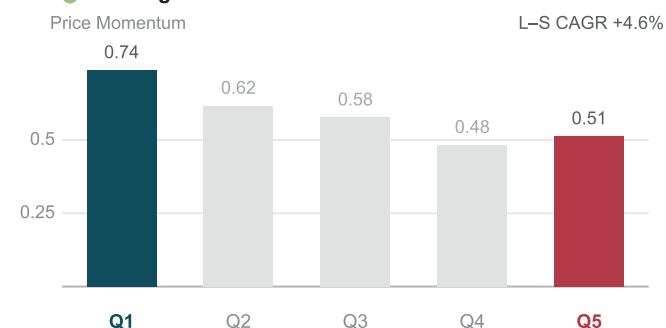


161 ● Num of Quarters of Declines in Receivables and Inventories/...



162 ● Dividend Growth

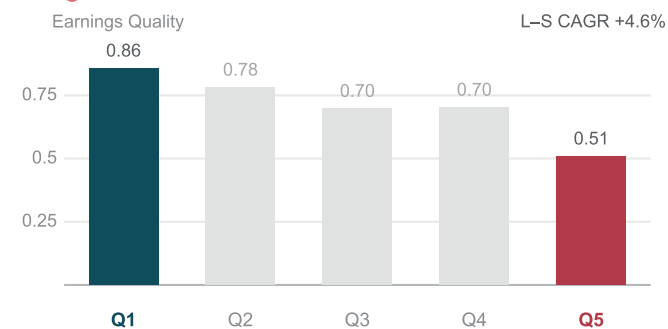


163 ● **Asset Turnover**164 ● **Share Turnover**165 ● **Adj Number of EPS FY2 Revisions**166 ● **Accrual Ratio - Cash Flows**167 ● **60 Day Coefficient of Variation of Volume to Price**168 ● **52W High Low**

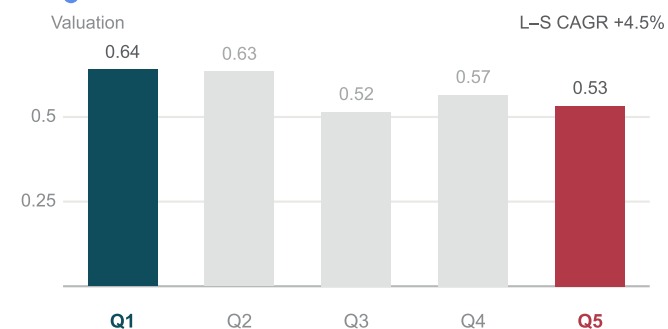
169 ● 1Y Chg in Sales Turnover



170 ● Current Ratio



171 ● Cash to Total Assets



172 ● 3Y Chg in Sales Turnover

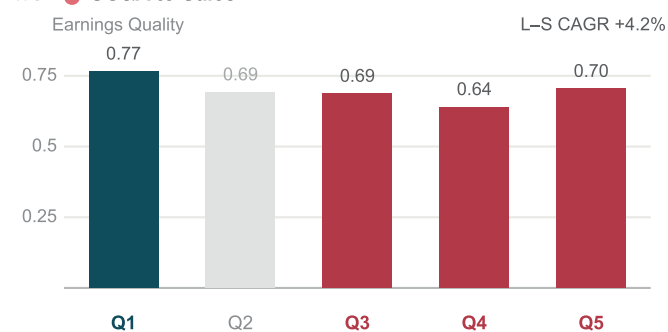
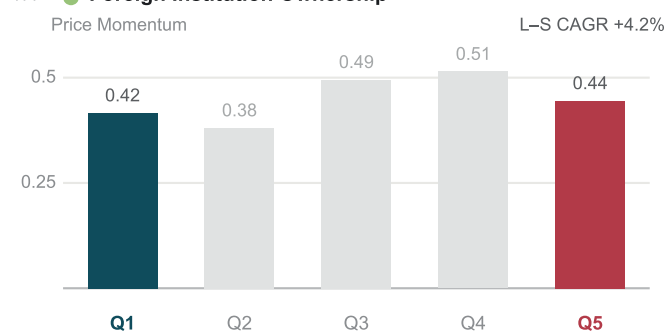


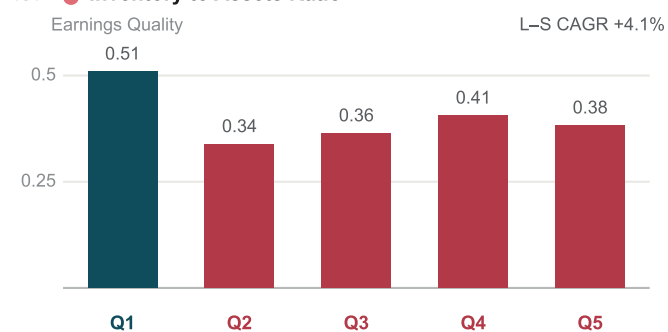
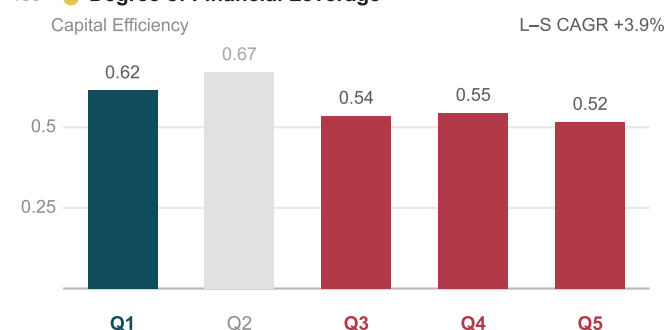
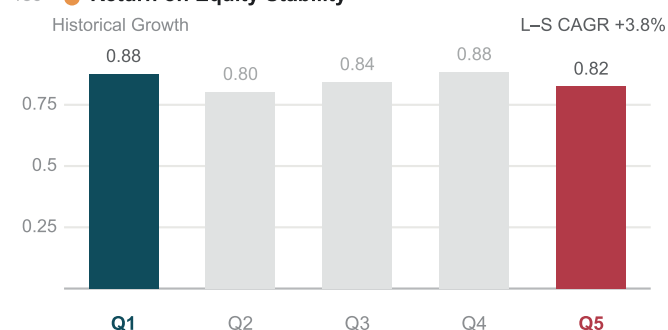
173 ● 6M Momentum in Trailing 12M Sales



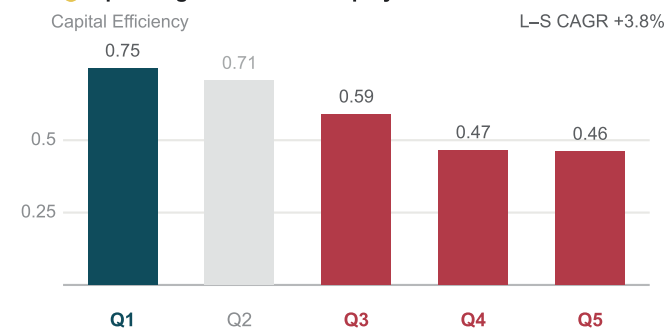
174 ● Change in Number of Holders



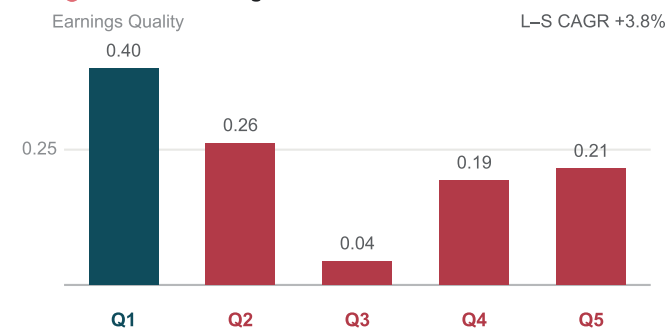
175 ● **3Y Chg in Asset Adjusted Free Cash Flow**176 ● **SG&A to Sales**177 ● **Foreign Institution Ownership**178 ● **Net Current Assets to Price Ratio**179 ● **Sales to Gross Profit Margin Growth**180 ● **Risk Adjusted Relative Strength**

181 ● **Inventory to Assets Ratio**182 ● **Log of Market Capitalization Cubed**183 ● **ROA 60 Month Slope**184 ● **Asset Adjusted Capital Investments**185 ● **Degree of Financial Leverage**186 ● **Return on Equity Stability**

187 ● Operating Cash Flow to Equity



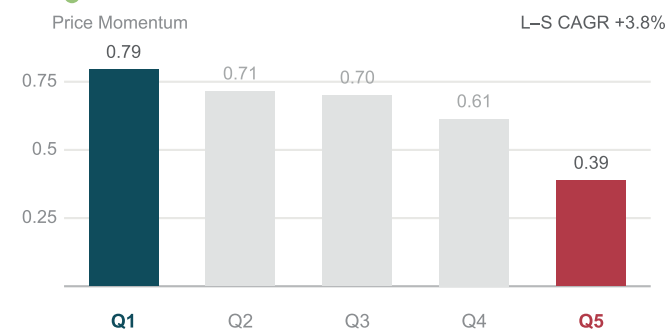
188 ● Retained Earnings to Total Assets



189 ● Net Profit Margin



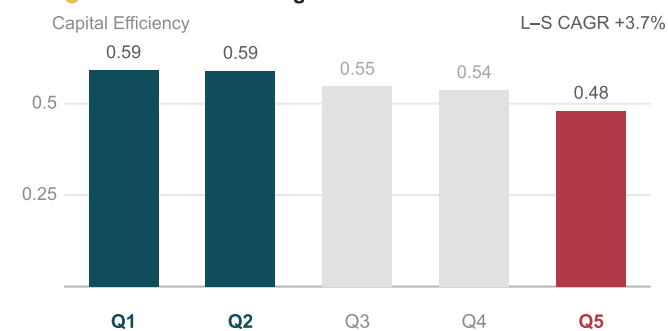
190 ● 26W Price Stat

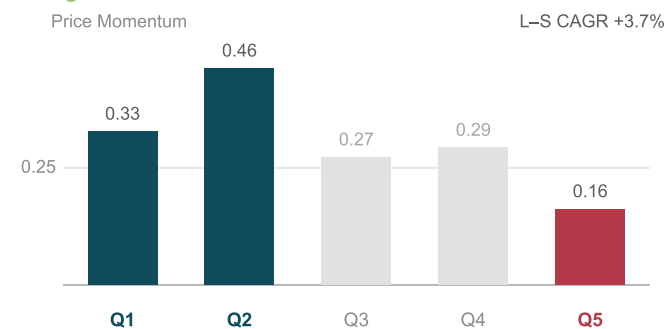
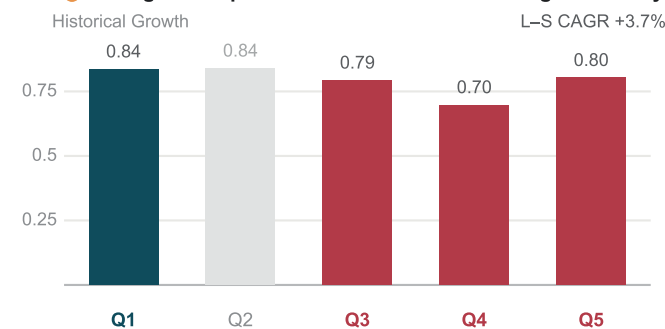
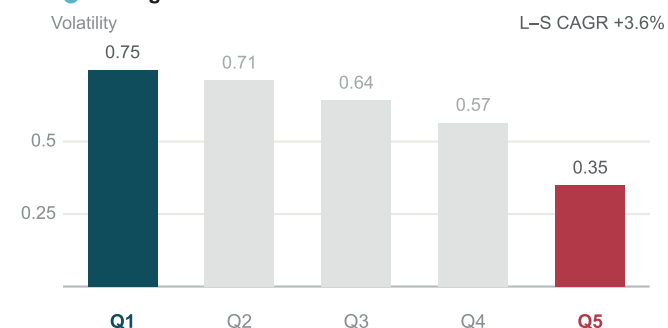
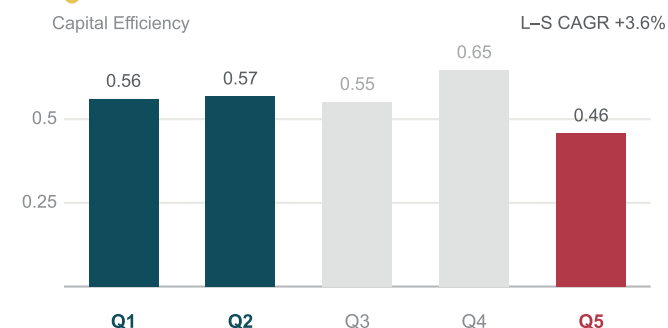


191 ● 3Y Chg in Asset Adjusted EPS

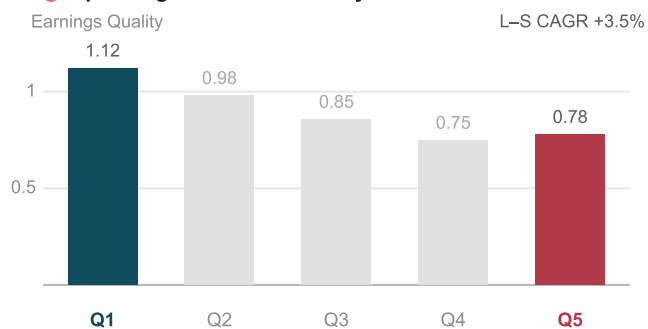


192 ● Year over Year Change of Total Debt

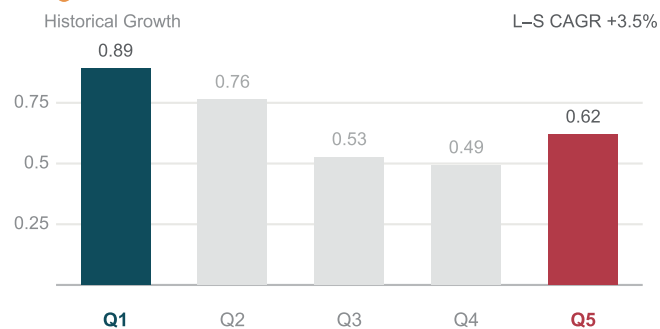


193 ● **24M Residual Return Variance**194 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**195 ● **Trailing 12-Month Overhead/Trailing 12-Month Sales**196 ● **Log Market Cap**197 ● **1Y Chg in Amihud**198 ● **Asset Growth - 1 Year**

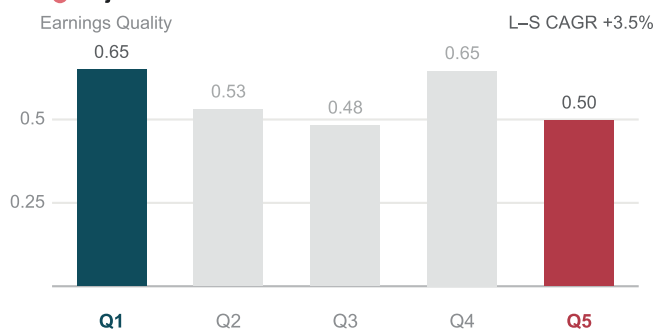
199 ● Operating Cash Flow Stability



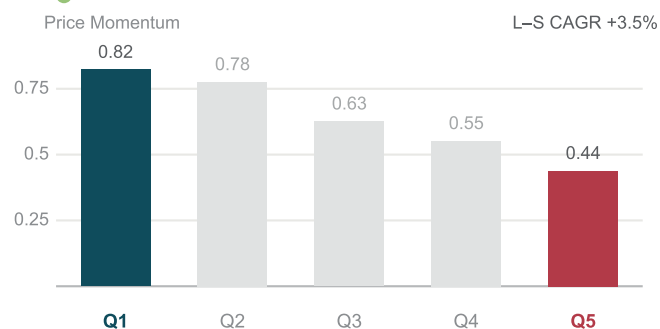
200 ● Sustainable Growth Rate



201 ● Adjusted Accruals



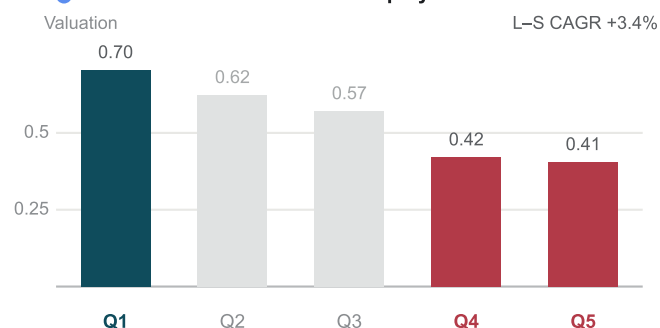
202 ● 180D Price TStat



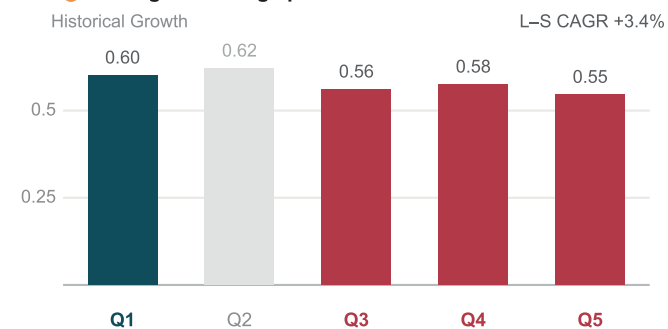
203 ● Target Price Gap to 6M EMA



204 ● Free Cash Flow to Common Equity



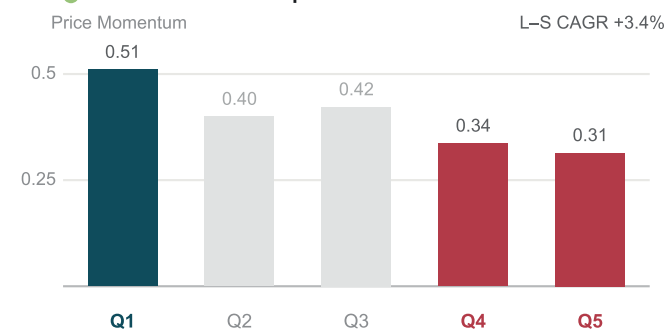
205 ● 1Y Chg in Earnings per Share



206 ● Working Capital to Total Assets



207 ● Institution Ownership Concentration



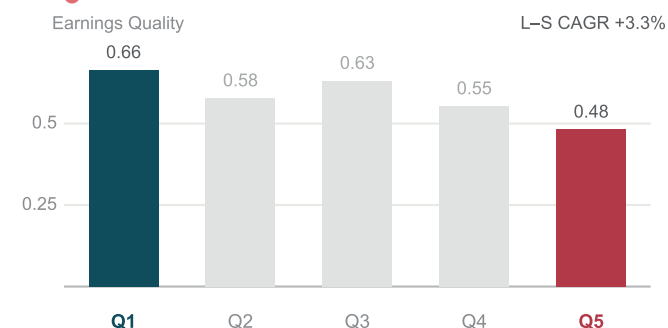
208 ● Return on Assets Stability



209 ● 3Y Chg in Earnings per Share



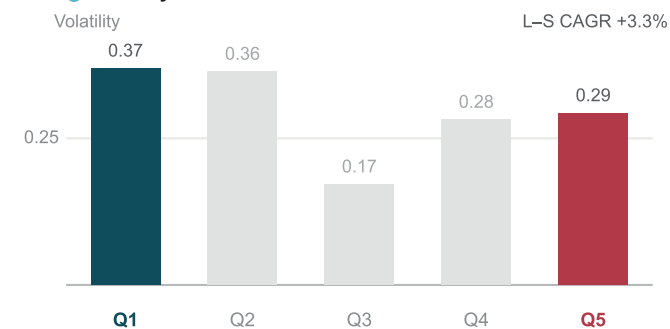
210 ● Accrual Ratio - Balance Sheet



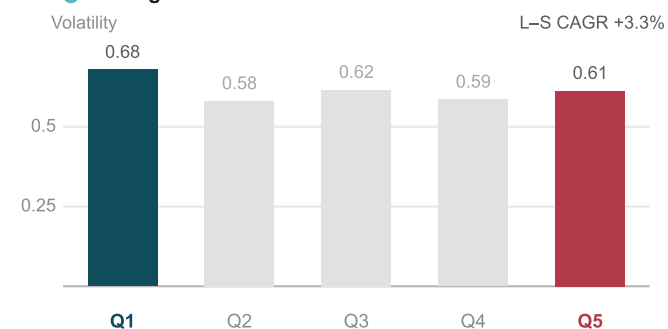
211 ● 3Y Chg in Operating Margin



212 ● 130 Day Minimum Return



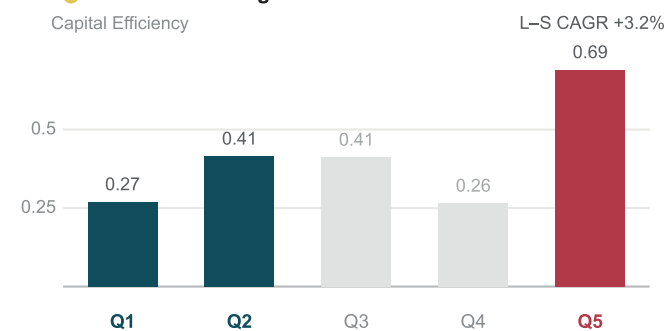
213 ● 1Y Chg in Debt to Assets



214 ● Ownership Breadth Stability of Institution Ownership by Buyers

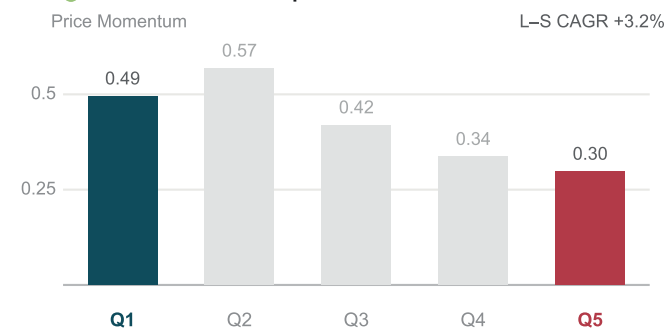
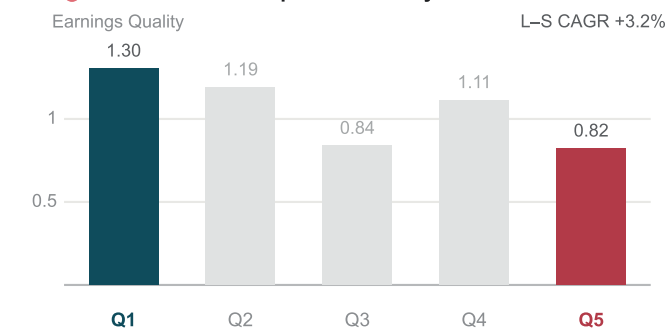
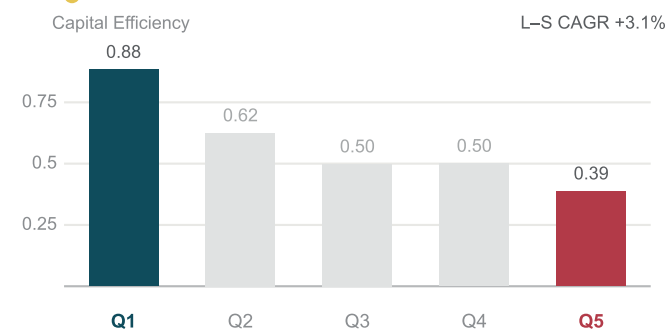


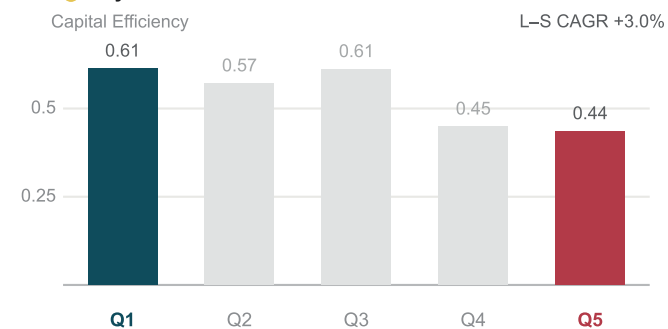
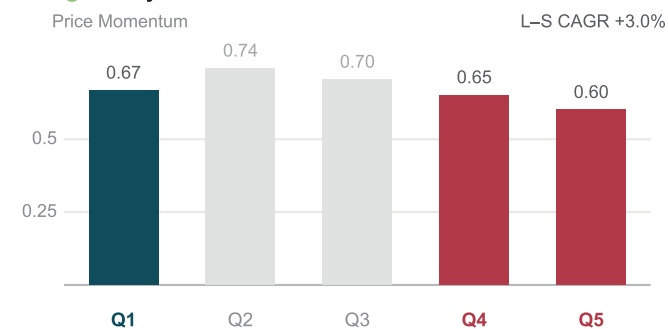
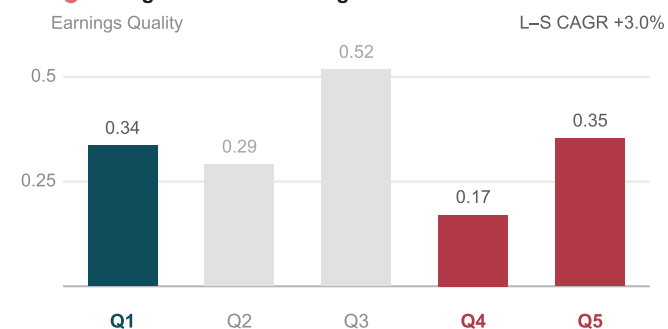
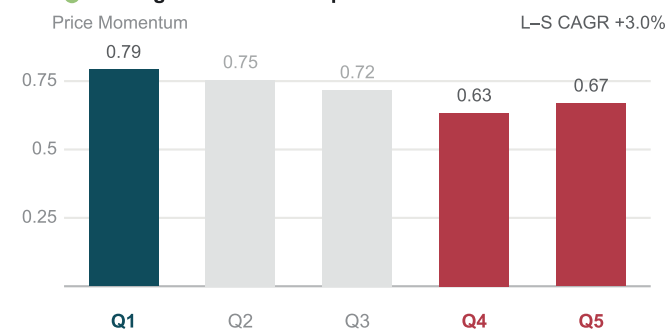
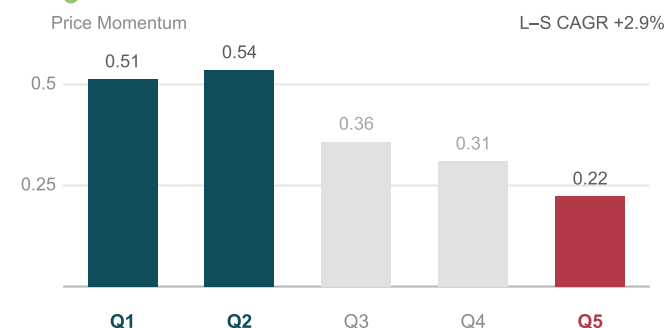
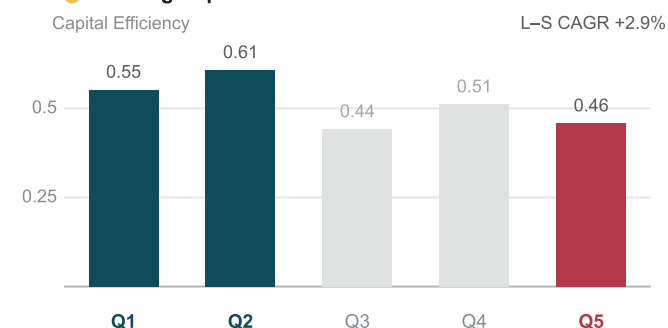
215 ● Pretax Profit Margin



216 ● 3Y Average Annual Sales Growth



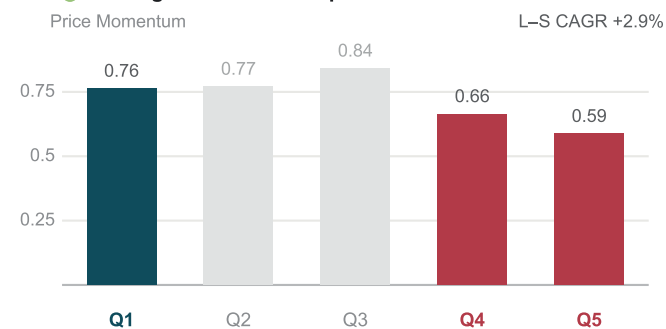
217 ● **Institution Ownership Level**218 ● **Research & Development Intensity**219 ● **1Y Chg in Sales to Earnings**220 ● **Gross Profit to Assets**221 ● **Cash Conversion Cycle**222 ● **Change in Number of Buyers**

223 ● **Payout Ratio**224 ● **10 Day MACD Trend**225 ● **Change of Sales to Change of EPS**226 ● **6M Chg in 18M CAPM Alpha**227 ● **Investment Duration**228 ● **Working Capital Turnover Ratio**

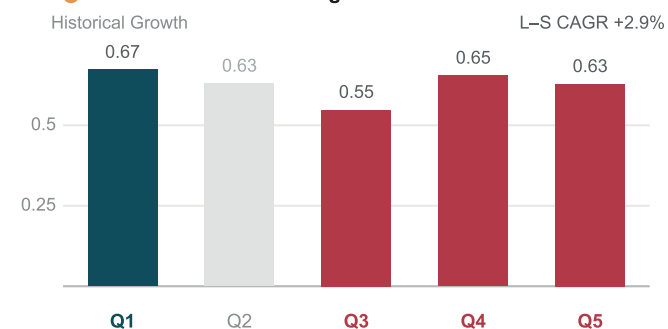
229 ● Ownership Breadth Stability of Institution Ownership by Hol...



230 ● 6M Chg in 36M CAPM Alpha



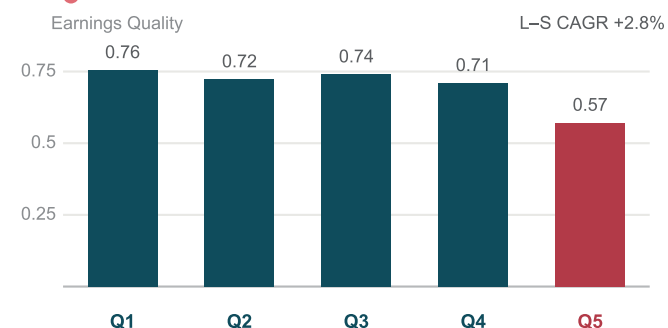
231 ● 3M Momentum in Trailing 12M Sales



232 ● Unexpected Receivables Change



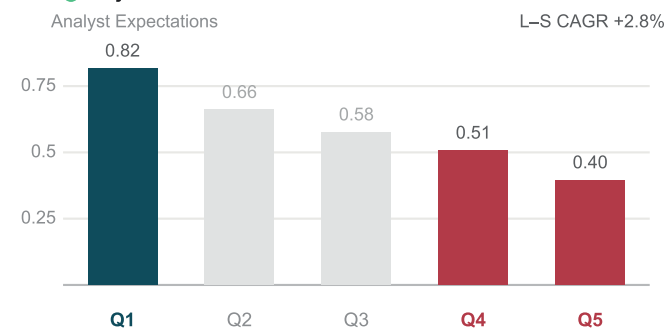
233 ● Cash to Sales



234 ● EBIT Estimate Dispersion



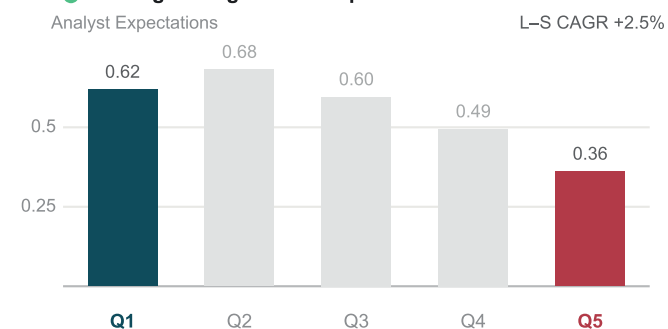
235 ● Adjusted Number of EPS FY1 Revisions



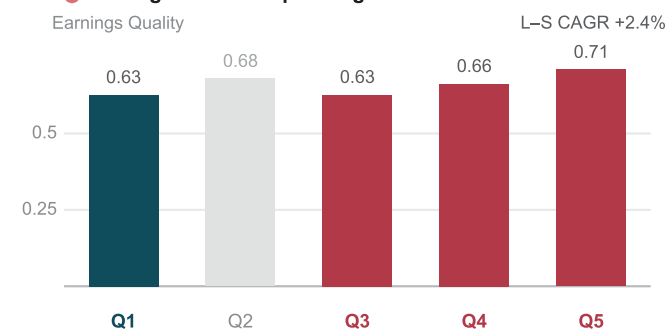
236 ● Volatility adjusted 12M return



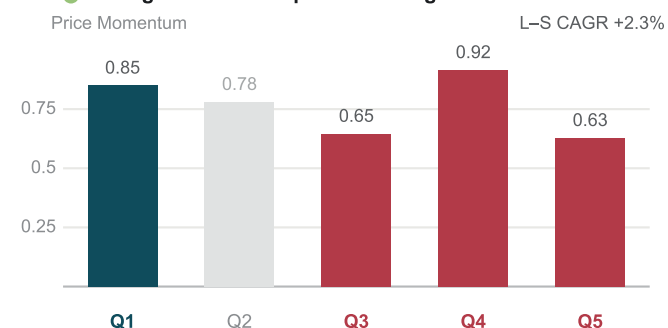
237 ● 6M Chg in Target Price Gap



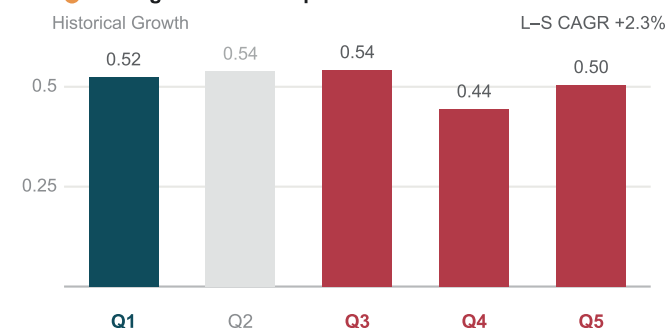
238 ● 1Y Chg in EPS to Operating Cash Flow

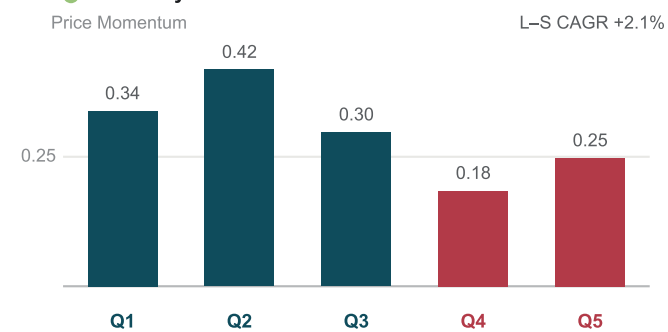
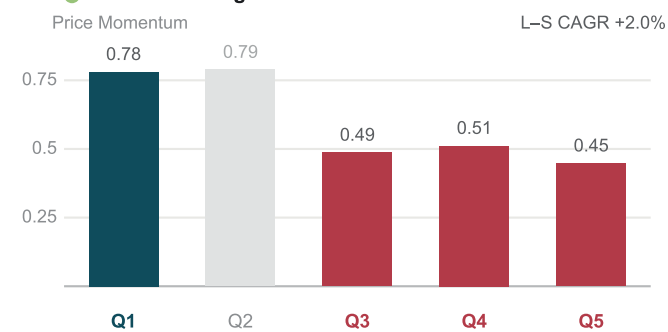
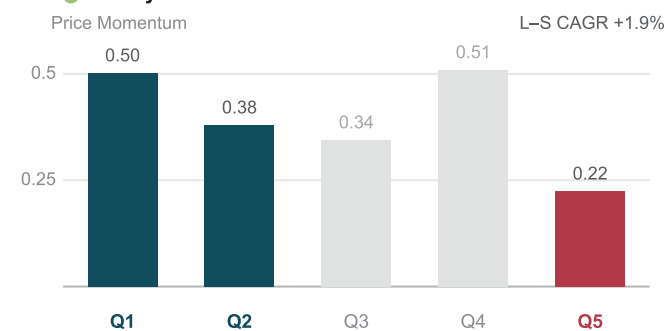
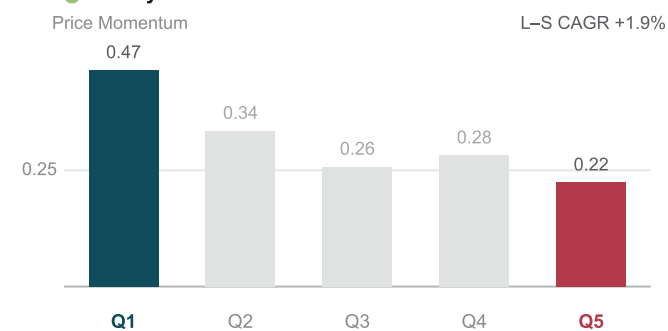


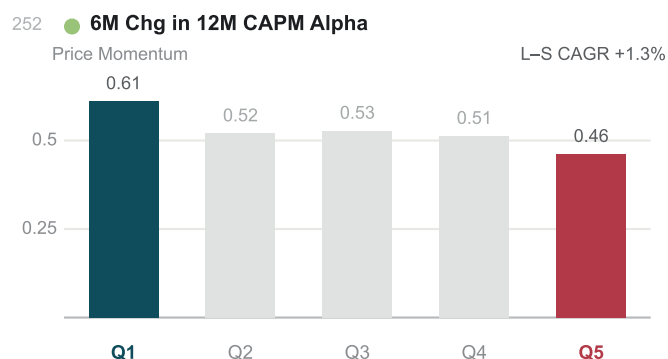
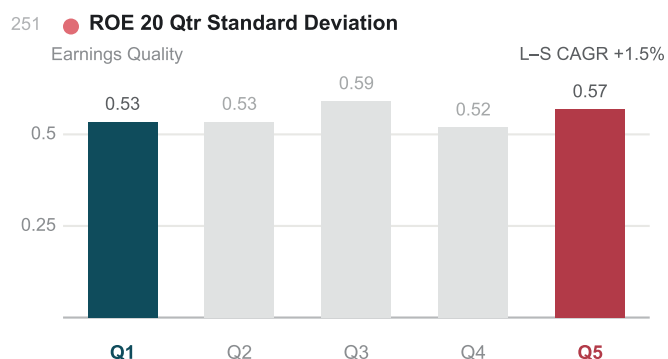
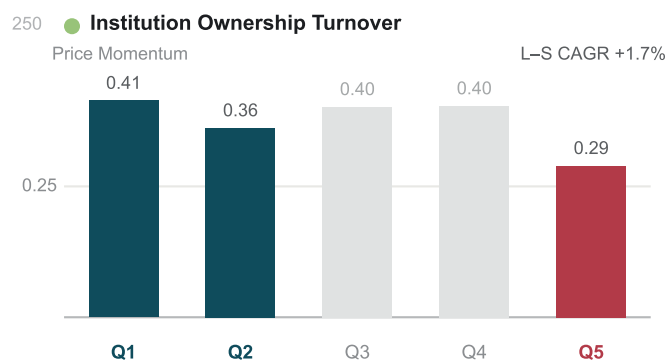
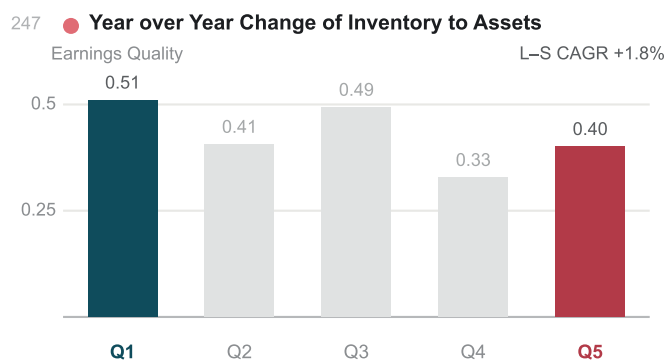
239 ● Change of Ownership Level - Hedge Fund



240 ● 1Y Chg in Cash Flow per Share



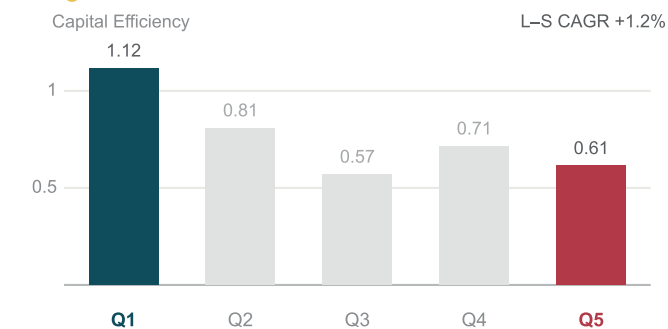
241 ● **Adj 50 Day Volume Signal**242 ● **Analyst Earnings Estimate Diffusion**243 ● **Max Daily Return in the Past 6 Months**244 ● **6 Month Average Share Turn Over**245 ● **30 Day Coefficient of Variation of Volume to Price**246 ● **20 Day Coefficient of Variation of Volume to Price**



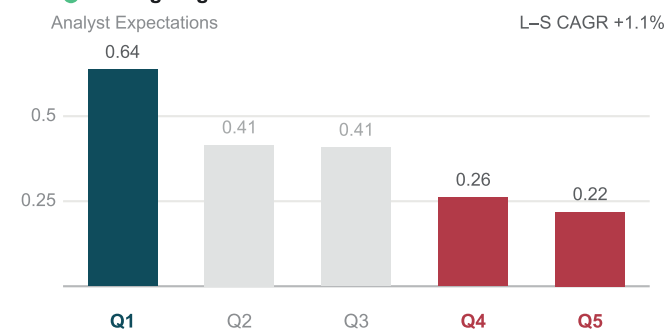
253 ● Sales Acceleration



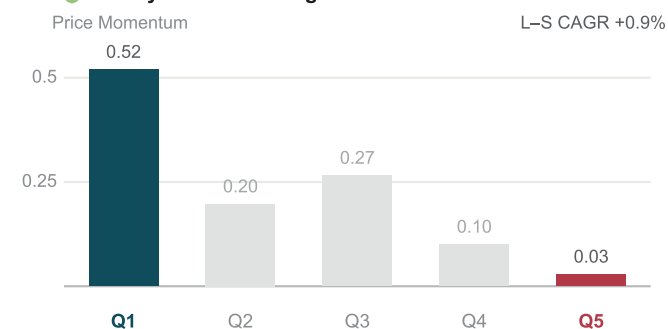
254 ● Normalized ROA



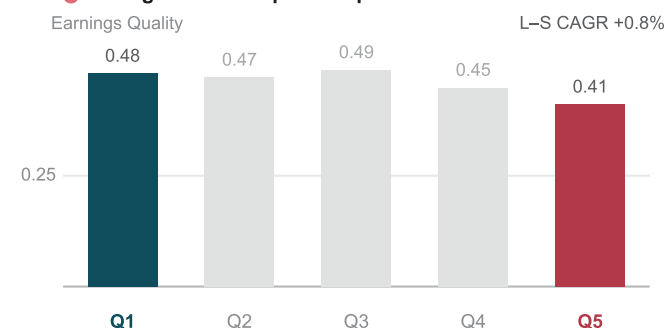
255 ● 6M Avg Chg 1M Recommendation



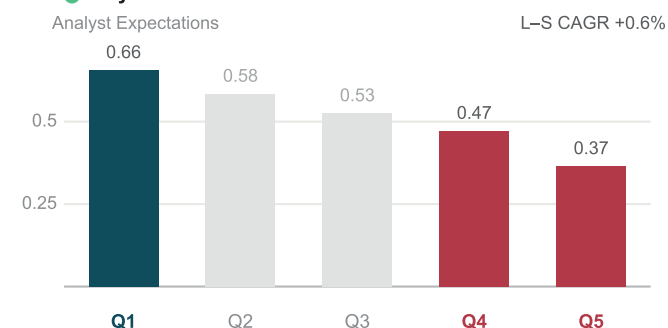
256 ● 14-Day Relative Strength Index



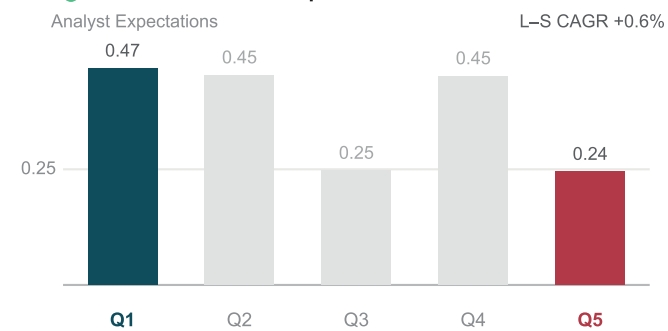
257 ● Change in TTM Depr. to Capex



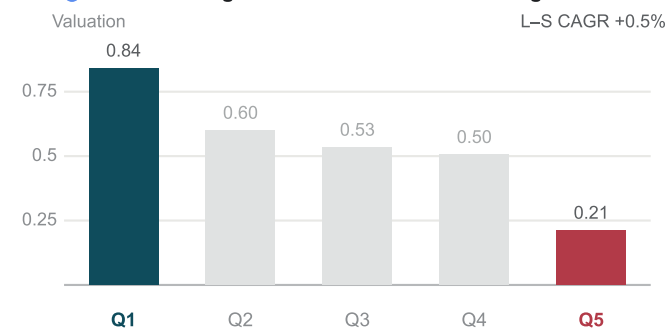
258 ● Buy-to-Sell Recommendation Ratio Less 3M EMA



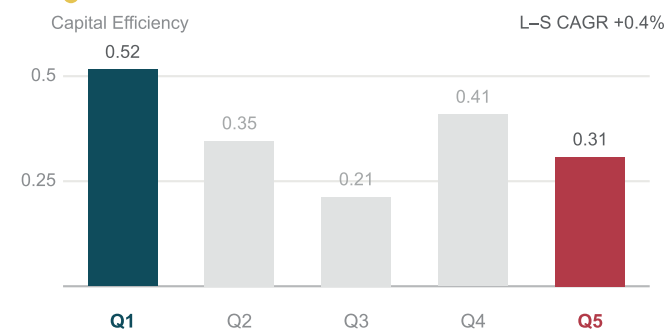
259 ● Revenue Estimate Dispersion



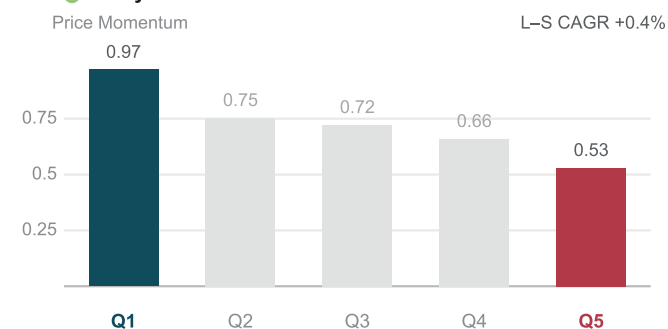
260 ● 8-Week Change in 12-Month Forward Earnings Consensus Estima...



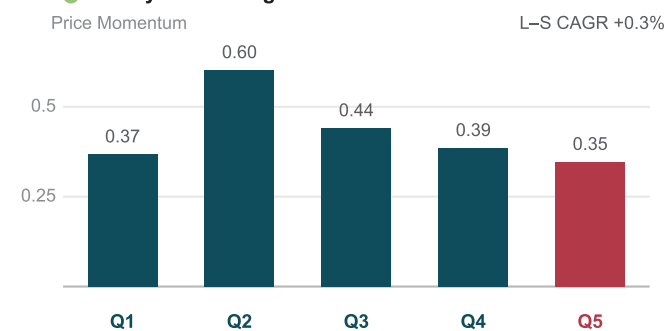
261 ● Normalized ROE



262 ● 5 Day Price Reversal



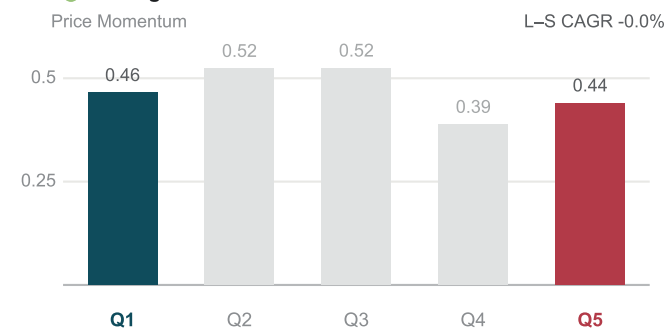
263 ● 50 Day Volume Signal



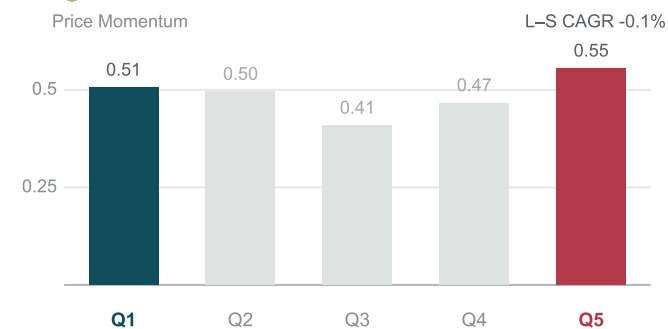
264 ● Change of Ownership Level - All Managers



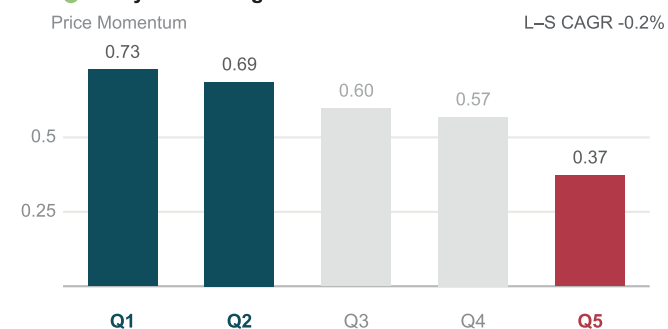
265 ● 1Y Chg in Share Turnover



266 ● 6 Months Price Reversal



267 ● 5 Day Volume Signal



268 ● 5-Day Money Flow/Volume



269 ● 4-Week Change in 12-Month Forward Earnings Consensus Estima...



270 ● 20-Day Lane's Stochastic Indicator

